



中国科学院计算技术研究所
INSTITUTE OF COMPUTING TECHNOLOGY, CHINESE ACADEMY OF SCIENCES



北京开源芯片研究院
BEIJING INSTITUTE OF OPEN SOURCE CHIP



香山开源处理器社区
XiangShan Open-Source Processor Community

XiangShan: An Open-Source High-Performance RISC-V Processor and Infrastructure for Architecture Research

The XiangShan Team
MICRO 25 @ Seoul, Korea
October 19, 2025

Opening Remarks

The Era of Open-Source Chip



A chip design that changes everything

- 10 Breakthrough Technologies 2023

Ever wonder how your smartphone connects to your Bluetooth speaker, given they were made by different companies? Well, Bluetooth is an open standard, meaning its design specifications, such as the required frequency and its data encoding protocols, are publicly available. Software and hardware based on open standards—Ethernet, Wi-Fi, PDF—have become household names.

Now an open standard known as RISC-V (pronounced “risk five”) could change how companies create computer chips.

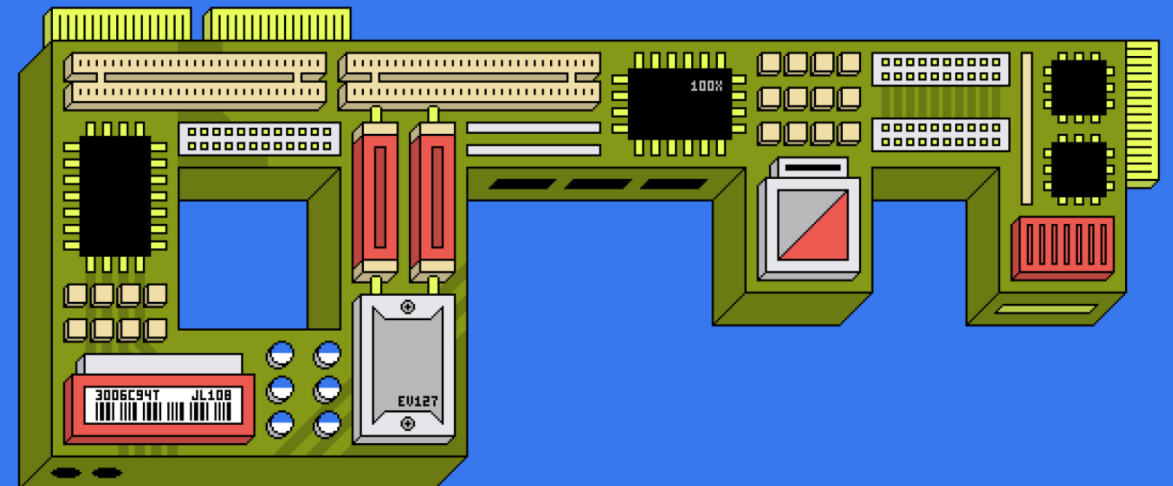
--- MIT Technology Review

A chip design that changes everything: 10 Breakthrough Technologies 2023

Computer chip designs are expensive and hard to license. That's all about to change thanks to the popular open standard known as RISC-V.

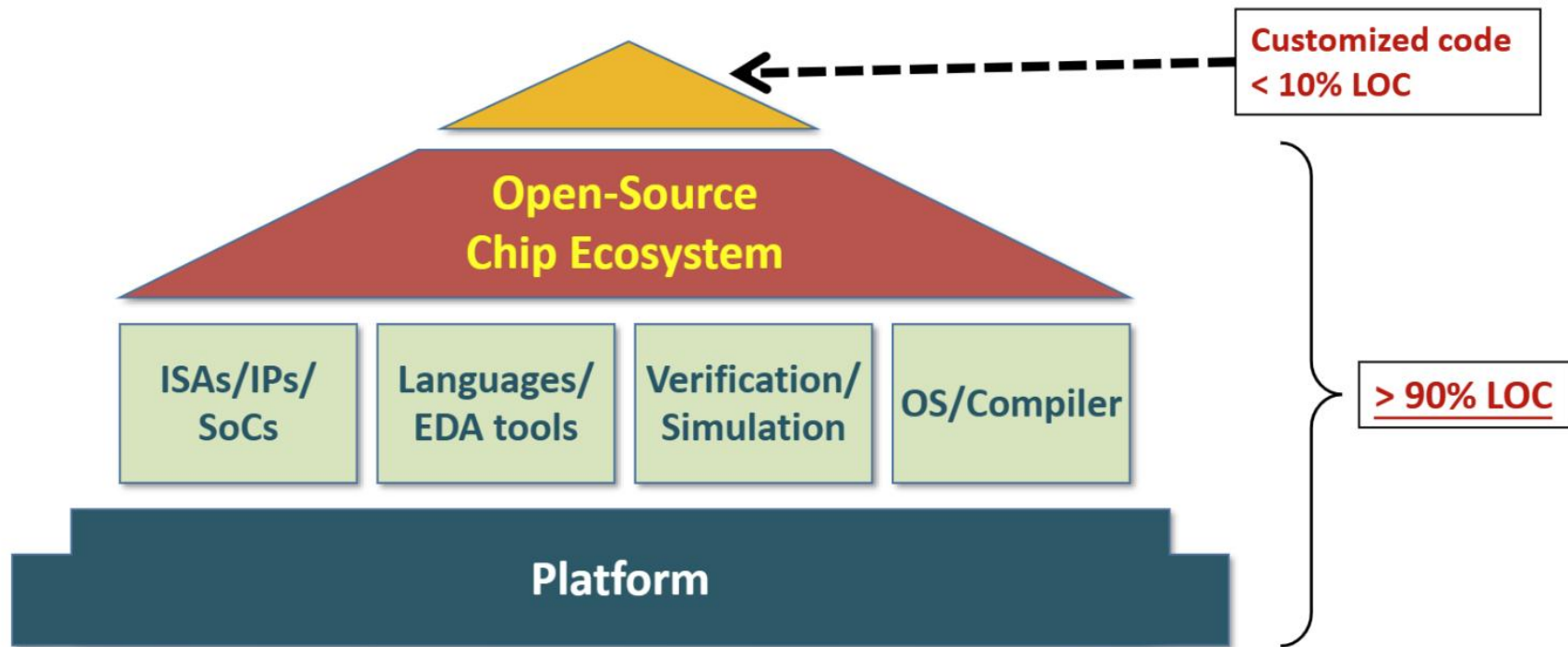
By Sophia Chen

January 9, 2023



🏔️ Open-Source Chip Ecosystem (OSCE)

- Goal: mirror the success of the open-source software ecosystem



To Lower the barrier of chip development

By saving the cost of IPs, EDA tools and engineers in chip design



Three Levels of Open-Source Chip

L1: OPEN ISA

L2: OPEN Design/Implementation

L3: OPEN Framework/Tools

Open Framework

Open Flow

Open Tools

microarch.

coding

EDA Tools

ISA Spec.

Docs

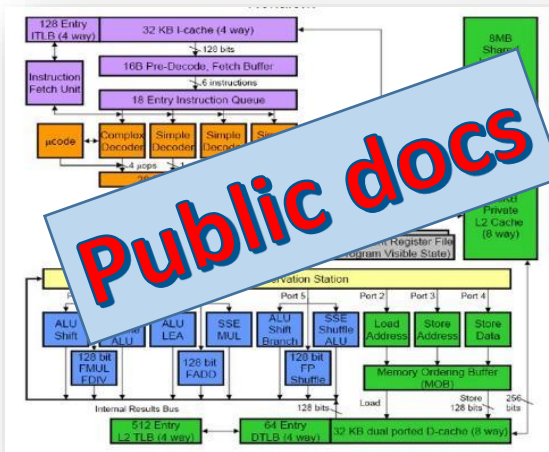
RTL

Layout

Open & Free

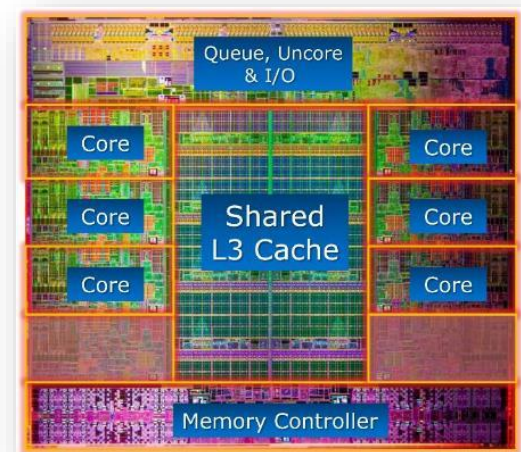
Public docs

Open-source



```
component DebugCoreTop is
port (
  -- Trigger and Data
  cu_Clk      : in  std_logic_vector(10 downto 0);
  cu0_Trig    : in  t_trig_0 := '0';
  cu1_Trig    : in  t_trig_1 := '0';
  cu2_Trig    : in  t_trig_2 := '0';
  cu0_Data    : in  std_logic_vector(31 downto 0);
  cu1_Data    : in  std_logic_vector(31 downto 0);
  cu2_Data    : in  std_logic_vector(31 downto 0);

  gt_RX_p     : in  std_logic := '0';
  gt_RefClk_n : in  std_logic := '0';
  gt_RX_n     : in  std_logic_vector(2 downto 0) := (others => '0');
  gt_TX_p     : out std_logic_vector(2 downto 0);
  gt_TX_n     : out std_logic_vector(2 downto 0);
);
end component;
```





Three levels of OSCE

L1: OPEN ISA

L2: OPEN Design & Implementation

L3: OPEN Tools & Infrastructure

3 Open Tools & Infrastructure

Microarch.

Engineering

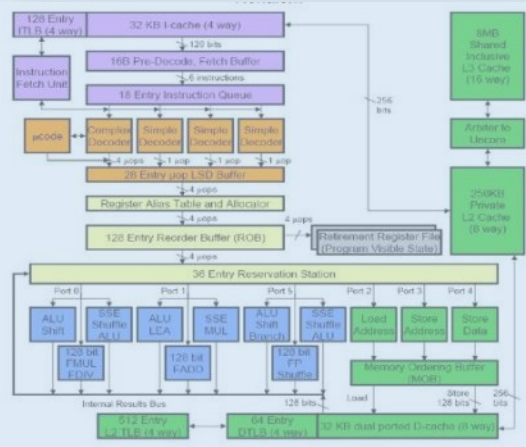
EDA Tools

ISA Spec.



1 Open ISA

Docs



2 Open Design & Implt

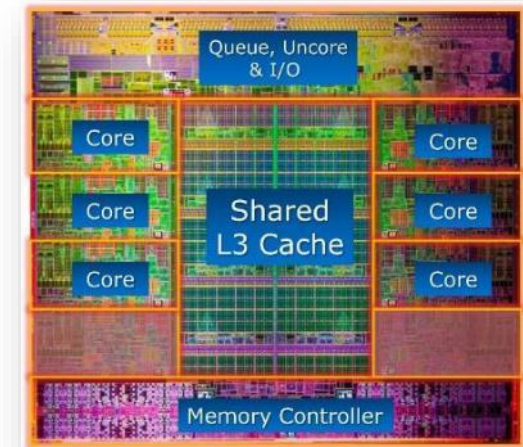
RTL codes

```
component DebugCoreTop is
port (
  -- Trigger and Data
  cu_Clk      : in   std_logic_vector(2 downto 0) := (others => '0');
  cu0_Trig    : in   t_trig_0 := (others => (others => '0'));
  cu1_Trig    : in   t_trig_1 := (others => (others => '0'));
  cu2_Trig    : in   t_trig_2 := (others => (others => '0'));
  cu0_Data    : in   t_data_0 := (others => (others => '0'));
  cu1_Data    : in   t_data_1 := (others => (others => '0'));
  cu2_Data    : in   t_data_2 := (others => (others => '0'));

  -- Downstream I2C
  SCL         : in   std_logic := '0';
  SDA         : inout std_logic := '0';

  -- Upstream
  gt_RefClk_p : in   std_logic := '0';
  gt_RefClk_n : in   std_logic := '0';
  gt_RX_p     : in   std_logic_vector(2 downto 0) := (others => '0');
  gt_RX_n     : in   std_logic_vector(2 downto 0) := (others => '0');
  gt_TX_p     : out  std_logic_vector(2 downto 0);
  gt_TX_n     : out  std_logic_vector(2 downto 0)
);
end component;
```

Layout





XiangShan: Open-Source High Performance Processors



L1: OPEN ISA

L2: OPEN Design/Implementation

L3: OPEN Framework/Tools



Fragrant Hill in Beijing

XiangShan Public

Edit Pins Unwatch 97 Fork 823 Starred 6.7k

master 295 Branches 5 Tags

Go to file Add file Code

About

Open-source high-performance RISC-V processor

xiangshan.cc

chisel risc-v microarchitecture

Readme View license Activity Custom properties 6.7k stars 97 watching 823 forks Report repository

feat(CLINT): add new clint to match 1:1 ration between CLIN...	2 weeks ago
submodule(ChiselAIA): bump ChiselAIA to fix lose interrupt (...)	5 months ago
build: bump chisel 6.7.0 (#5069)	2 weeks ago
bump difftest & mkdir for wave/perf for local-ci script's run-...	2 years ago
fix(difftest): bump and align parseArg return values (#5009)	last month
chore: bump chisel 6.0.0 (#2654)	last year
build: bump chisel 6.7.0 (#5069)	2 weeks ago
docs(readme): update kunminghu architecture graph (#3910)	11 months ago
docs(readme): update kunminghu architecture graph (#3910)	11 months ago

> 6.7K stars, > 820 forks on GitHub



Why Open-Source High-perf. RISC-V Processor XiangShan?

- **Why RISC-V:** Free and open ISA
- **Why high-perf :** Most RISC-V processors are for IoT/edge, but both academic and industrial community need high-performance RISC-V processors
- **Why open-source:** An open and innovative hardware platform
- **Build a leading platform with end-to-end agile development flows and tools**



Linux

V.S.



XIANGSHAN

**Envision:
"Hardware Version of Linux"**



OpenXiangShan: Empowering Architecture Research

- An open-source, continuously developing research platform

Your research field	XiangShan provides ...
Microarchitecture	Performance: An industrial-competitive, high-performance superscalar OoO microarchitecture
	Functionality: RVA23-compatible RISC-V design
	Development: mature, user-friendly design flows
	Tapeouts by the XiangShan team and leading industrial partners with real-world deployment
Chip development tool	Realistic and challenging research problems



15+ XiangShan Tutorials Around the World

- HPCA, Edinburgh, Scotland
- ASPLOS, San Diego, USA
- RVSC, Hangzhou, China
- MICRO, Austin, USA

2024

- HPCA, Sydney, Australia
- More coming ...

2026

2023

- ASPLOS, Vancouver, Canada
- RVSC, Beijing, China
- MICRO, Toronto, Canada

2025

- HPCA, Las Vegas, USA
- ASPLOS, Rotterdam, Netherlands
- RVSE, Paris, France
- ISCA, Tokyo, Japan
- APPT, Athens, Greece
- RVSC, Shanghai, China
- **MICRO, Seoul, Korea**

***Welcome old
and new friends!***



What We Will Cover in This Tutorial

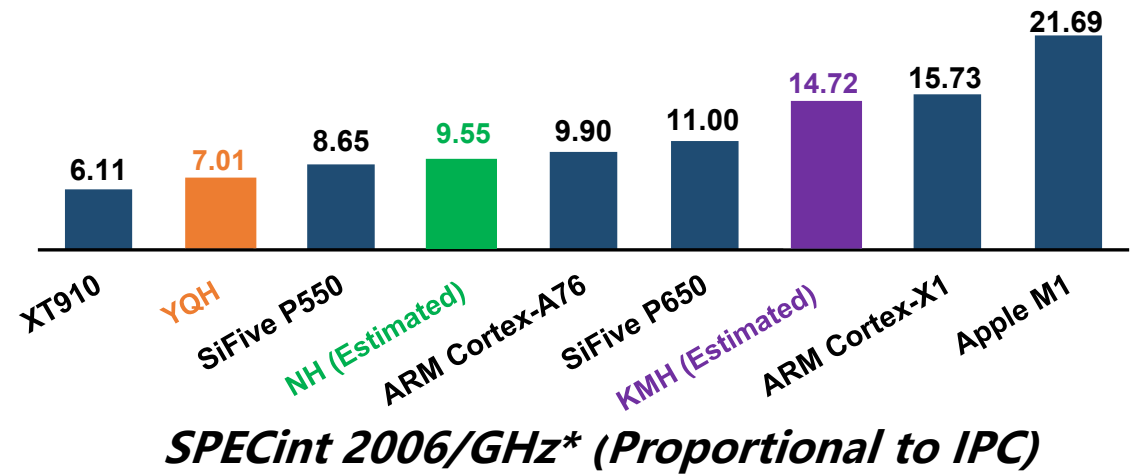
- General introduction (8:00 – 8:40, 40 minutes)
- Introduction of XS-Gem5 (8:40 – 10:00, 80 minutes)
- Coffee Breaks (10:00 – 10:30, 30 minutes)
- Microarchitecture design philosophy (10:30 – 11:00, 30 minutes)
- Development workflows (11:00 – 12:00, 60 minutes)

Open-Source, High Performance, Industrial Level RISC-V Processor



XiangShan: Open-Source High Performance Processors

- **1st generation: Yanqihu (YQH)**
 - RV64GC, single-core, superscalar OoO
 - 28nm tape-out, 1.3GHz, July 2021
 - SPEC CPU2006 7.01@1GHz, DDR4-1600
- **2nd generation: Nanhu (NH)**
 - RV64GCBK, dual-core, superscalar OoO
 - 14nm GDSII delivery, 2GHz, 2023 Q3
 - Estimated** SPECint 2006 19.10@2GHz
- **3rd generation: Kunminghu (KMH)**
 - RV64GCBKHV, quad-core, superscalar OoO
 - Advanced-node, 3GHz, 1.5x IPC of NH
 - Close collaboration with industrial partners



* Source: XT910@ISCA' 20, SiFive, AnandTech

** Updated January 5, 2023

XiangShan Gen 1: Yanqihu

- **Test chip developed almost entirely by students**
 - RV64GC, 11-stage, superscalar, out-of-order
 - 5.3 CoreMark/MHz (gcc-9.3.0 -O2)
- **Tape-out:** single XiangShan core with 1MB L2 Cache
 - 28nm, 1.3GHz, July 2021



Yanqi Lake in Beijing

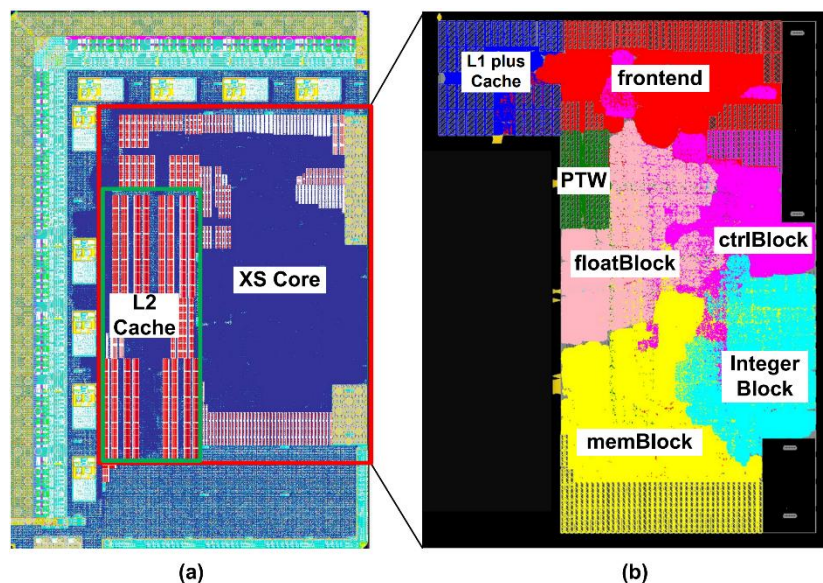


Figure. Layout of (a) the entire chip; (b) the core



Tape-out information for the processor core

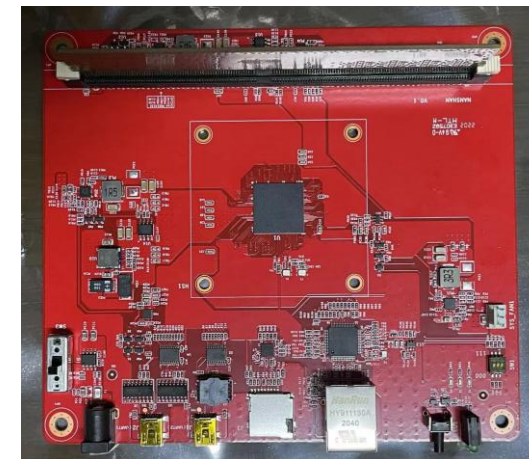
Process Node	28nm
Die Size	8.6 mm ²
Std Cell	5.05M, 4.27 mm ²
Mem	261, 1.7mm ²
Density	66%
Cell	ULVT 1.04%, LVT 19.32%, SVT 25.19%, HVT 53.67%
Estimated Power	5W
Frequency	1.3GHz, TT85C

Real Chip of XiangShan Gen 1 Yanqihu

- The chip was back in January 2022
 - SoC: CPU, SPI Flash, UART, SD card, Ethernet, DIMM
 - Correctly running Debian with SD card and ethernet
- Performance: SPEC CPU2006 7.01@1GHz

SPECint 2006 @ 1GHz		SPECfp 2006 @ 1GHz	
400.perlbench	6.14	410.bwaves	9.28
401.bzip2	4.37	416.gamess	6.59
403.gcc	6.71	433.milc	8.41
429.mcf	6.83	434.zeusmp	7.65
445.gobmk	7.92	435.gromacs	4.99
456.hmmer	5.24	436.cactusADM	3.97
458.sjeng	6.85	437.leslie3d	6.93
462.libquantum	17.71	444.namd	8.00
464.h264ref	10.91	447.dealII	10.17
471.omnetpp	5.65	450.soplex	7.03
473.astar	5.16	453.povray	7.14
483.xalancbmk	7.35	454.Calculix	2.86
		459.GemsFDTD	8.35
		465.tonto	6.42
		470.lbm	10.39
		481.wrf	7.26
		482.sphinx3	9.07

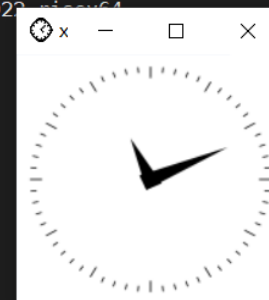
SPECint 2006: 7.03@1GHz
SPECfp 2006: 7.00@1GHz



```
wanghuizhe@open02:~$ ssh -X xs@172.28.2.246
xs@172.28.2.246's password:
Linux open02 4.20.0-44668-ge9c195ab0c63-dirty #109 Thu Feb 17 17:41:13 CST 2022; root:x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
You have no mail.
Last login: Thu Feb 17 11:10:31 2022 from 172.28.9.102
xs@open02:~$ xclock
Warning: locale not supported by C library, locale unchanged
```



**SSH into the Debian on
XiangShan, and
run a GUI program via X11
forwarding**

XiangShan Gen 2: Nanhu

- **Target: 2GHz@14nm, SPEC CPU2006 ~20 marks**
- **New frontend:** decoupled BP and instruction fetch
- **Improved backend:** better scheduler, instruction fusions, move elimination, and more
- **New L2/L3 cache:** designed for high frequency and high performance with hybrid prefetchers
- **Verified under dual-core config** (RV64GCBK), more devices support (PCIe, USB, ...)

Setup an open and standardized development workflow



A lake in Jiaying, Zhejiang, China

☐		10 Open	✓	2,322 Closed
☐		Bump diffTest for palladium simulation ✓		
		#2662 opened 11 hours ago by klin02	Approved	
☐		ICache: fix ICacheMainPipe bug about sfence ✓	bug	do not merge
		#2660 opened 3 days ago by ssszwic	Review required	
☐		ICache: change data SRAM partWayNum from 2 to 4 ✓		
		#2653 opened 5 days ago by ssszwic	Approved	
☐		ITTAGE meta width shrink ✓	do not merge	
		#2592 opened last month by eastonman	Review required	
☐		pf: fix negative stream ✓	bug	
		#2508 opened on Nov 27, 2023 by happy-lx	Review required	
☐		SQ: add sq merge ✓	do not merge	enhancement
		#2439 opened on Oct 29, 2023 by sfencevma	Review required	
☐		rm refillPipe ✓	enhancement	
		#2426 opened on Oct 25, 2023 by YukunXue	Approved	

Pull Request
Snapshot

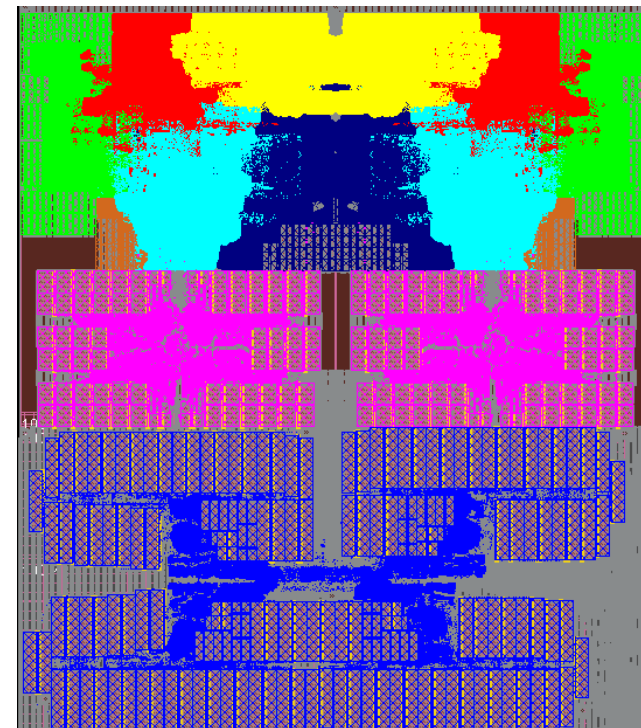


Estimated Performance of XiangShan Gen 2 Nanhu

- Estimated **SPECint 2006 19.10, SPECfp 2006 22.18@2GHz**
 - RTL simulation, DDR4-2400 under DRAMsim3
 - Compile with GCC 10.2.0, -O2, RV64GCB

SPECint 06	
400.perlbench	19.27
401.bzip2	11.36
403.gcc	21.97
429.mcf	20.53
445.gobmk	15.97
456.hmmer	19.22
458.sjeng	17.22
462.libquantum	36.99
464.h264ref	28.54
471.omnetpp	14.02
473.astar	14.19
483.xalancbmk	21.48
SPECint2006@2GHz	19.10

SPECfp 06	
410.bwaves	18.09
416.gamess	23.82
433.milc	18.33
434.zeusmp	28.19
435.gromacs	17.53
436.cactusADM	24.26
437.leslie3d	20.28
444.namd	23.83
447.dealII	33.50
450.soplex	25.61
453.povray	27.06
454.Calculix	9.18
459.GemsFDTD	24.66
465.tonto	17.68
470.lbm	32.04
481.wrf	19.73
482.sphinx3	28.38
SPECfp2006@2GHz	22.18



Feature	V1YQH	V2NH
ISA	RV64GC	RV64GCBK
Process Node	28nm	14nm
Core Count	1	2
Die Size	8.6 mm ²	22.13 mm ²
Std Cell Num/Area	5.05M, 4.27 mm ²	11.3M, 4.53 mm ²
Mem Num/Area	261, 1.7 mm ²	692, 8.93 mm ²
Density	66%	35%
Frequency	1.3GHz, TT 0.9V	2GHz, TT 0.9V

* Updated January 5, 2023. Performance confirmed on FPGA as well

XiangShan Gen 3: Kunminghu

- **Target ARM Neoverse N2**
 - SPECCPU2006: 45@3GHz (15/GHz)
 - Vector/Hypervisor extension supported
- **A Joint Dev Team** (coordinated by BOSC)
 - About 10 institutions



Kunming Lake in Beijing





Highlights in XiangShan Gen 3 Kunminghu

- **Functional Enhancement**

- Support RISC-V **Vector/Hypervisor** extension
- Support **RVA23 profile**
- Support interconnection based on **CHI protocol**

- **Performance Exploration**

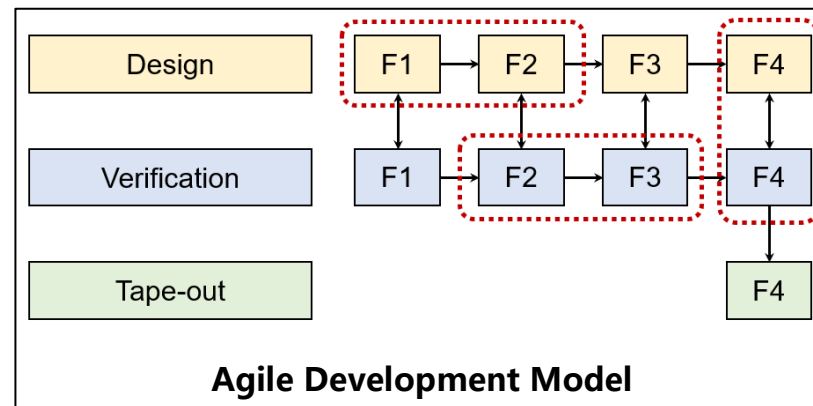
- Performance boost in frontend, backend, load-store unit and cache
- Performance model calibrated with RTL
- Workflow: **DSE on perf model => Impl. & fine tuning on RTL**

- **Functional Verification**

- **Hierarchical verification flow** spanning system/integration/unit level + FPGA prototyping
- Industrial-grade verification process

- **Physical Design**

- Experienced physical design team
- Simultaneous iteration of RTL coding based on timing evaluation





Compiler Support

- LLVM & GCC Now Include XiangShan Kunminghu V2R2 Optimizations

```
llvm / llvm-project

<> Code 5k+ Pull requests 4.4k Actions Security Insights

Commit 0e3e0bf
4 people authored 3 days ago · 21 / 23 · Verified

[RISCV] Add processor definition for XiangShan-KunMingHu-V2R2 (#123193)

XiangShan-KunMingHu is the third generation of Open-source
high-performance RISC-V processor developed by Beijing Institute of Open
Source Chip (BOSC), and its latest version is V2R2.

The KunMingHu manual is now available at
https://github.com/OpenXiangShan/XiangShan-User-Guide/releases.
It will be updated on the official XiangShan documentation site:
https://docs.xiangshan.cc/zh-cn/latest

You can find the corresponding ISA extension from the XiangShan Github
repository:
https://github.com/OpenXiangShan/XiangShan/blob/master/src/main/scala/xiangshan/Parameters.scala

If you want to track the latest performance data of KunMingHu, please
check XiangShan Biweekly: https://docs.xiangshan.cc/zh-cn/latest/blog

This PR adds the processor definition for KunMingHu V2R2, developed by
the XSCC team https://github.com/orgs/OpenXiangShan/teams/xscc.

The scheduling model for XiangShan-KunMingHu V2R2 will be submitted in a
subsequent PR.

-----

Co-authored-by: Shenglin Tang <tangshenglin@ict.ac.cn>
Co-authored-by: Xu, Zefan <ceba_robot@outlook.com>
Co-authored-by: Tang Haojin <tanghaojin@outlook.com>

main (#123193)
```

```
gcc-mirror / gcc

<> Code Pull requests 48 Actions Projects Wiki Security Insights

Commit f0cd40f
3 people committed on Jun 6

RISC-V: Support -mcpu for XiangShan Kunminghu cpu.

This patch adds support for the XiangShan Kunminghu CPU in GCC, allowing
the use of the '-mcpu=xiangshan-kunminghu' option.

XiangShan-KunMingHu is the third-generation open-source high-performance
RISC-V processor.[1] You can find the corresponding ISA extension from the
XiangShan Github repository.[2] The latest news of KunMingHu can be found
in the XiangShan Biweekly.[3]

[1] https://github.com/OpenXiangShan/XiangShan-User-Guide/releases.
[2] https://github.com/OpenXiangShan/XiangShan/blob/master/src/main/scala/xiangshan/Parameters.scala
[3] https://docs.xiangshan.cc/zh-cn/latest/blog

A dedicated scheduling model for KunMingHu's hybrid pipeline will be
proposed in a subsequent PR.

gcc/ChangeLog:

    * config/riscv/riscv-cores.def (RISCV_TUNE): New cpu tune.
    (RISCV_CORE): New cpu.
    * doc/invoke.texi: Ditto.

gcc/testsuite/ChangeLog:

    * gcc.target/riscv/mcpu-xiangshan-kunminghu.c: New test.

Co-Authored-By: Jiawei Chen <jiawei@iscas.ac.cn>
Co-Authored-By: Yangyu Chen <cyy@cyyself.name>
Co-Authored-By: Tang Haojin <tanghaojin@outlook.com>

master
```

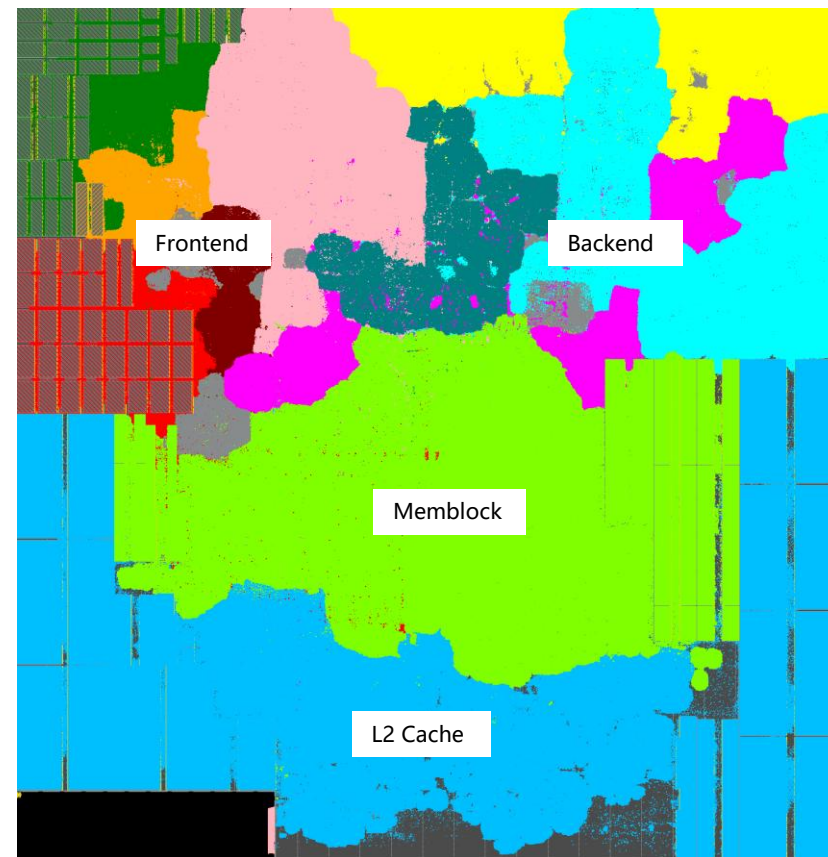


Performance Evaluation of Gen 3 Kunminghu

- Method: SPEC CPU checkpoints selected by Simpoint
 - Base: GCC 12 -O3, RV64GCB, jemalloc
 - 1MB L2 and 16MB L3
 - Simulated@3GHz with DRAMsim3 DDR4-3200

SPECint 2006 est.@ 3GHz		SPECfp 2006 est.@ 3GHz	
400.perlbench	35.88	410.bwaves	66.89
401.bzip2	25.50	416.gamess	40.89
403.gcc	46.72	433.milc	45.25
429.mcf	58.13	434.zeusmp	52.10
445.gobmk	30.26	435.gromacs	33.65
456.hmmer	41.60	436.cactusADM	46.16
458.sjeng	30.53	437.leslie3d	46.01
462.libquantum	122.50	444.namd	28.88
464.h264ref	56.57	447.dealII	73.43
471.omnetpp	39.37	450.soplex	51.99
473.astar	29.23	453.povray	53.44
483.xalancbmk	72.03	454.Calculix	16.38
GEOMEAN	44.15	459.GemsFDTD	37.18
		465.tonto	36.67
		470.lbm	91.24
		481.wrf	40.62
		482.sphinx3	48.57
		GEOMEAN	44.60

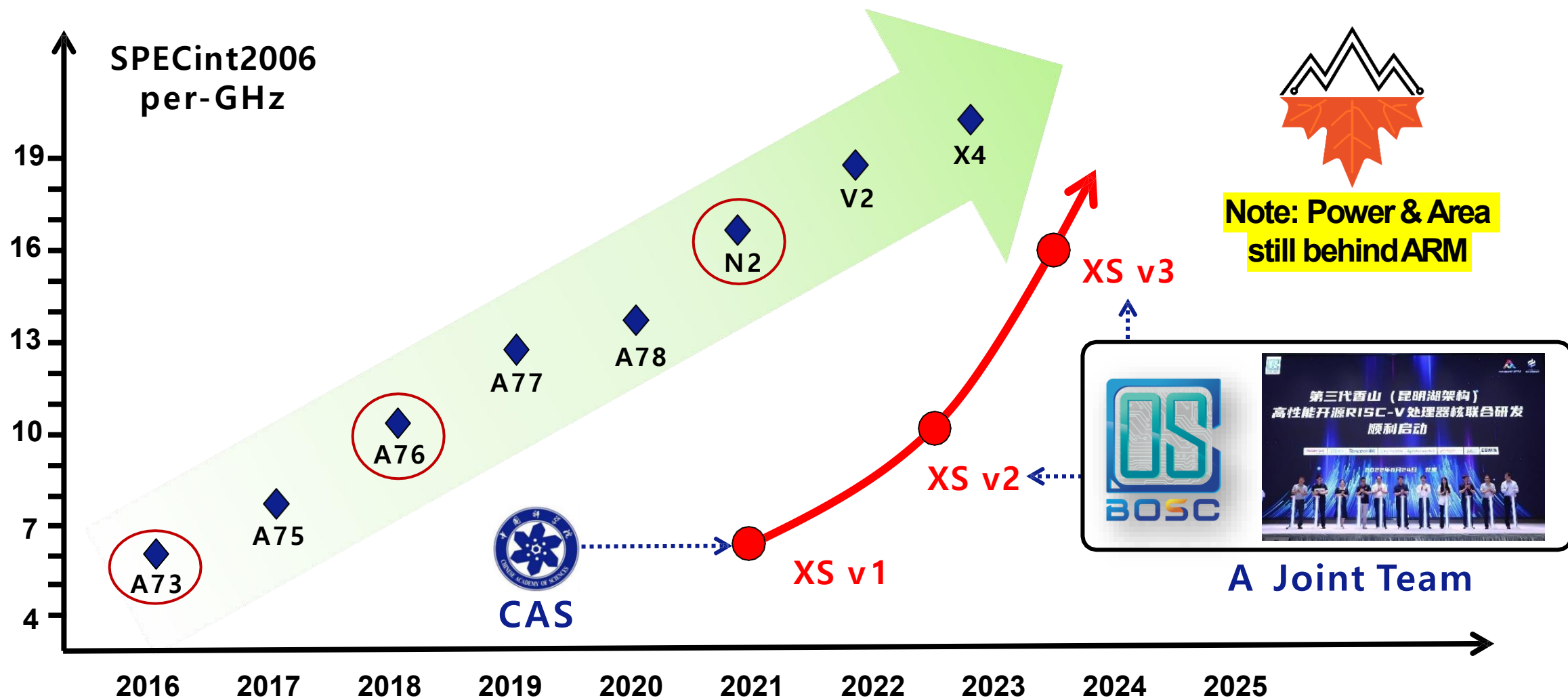
* Updated in March 2025



Floorplan of KMH V2R2 (single core)



XiangShan: Open-Source High Performance Processors



Prospect: "Dual Core" Roadmap



Based on **V3**
(KMH)

- **Big Core: Ultimate Performance** (v.s. **ARM N2**)

Target High-Throughput Advanced Computing Platform

Goal: become mainstream CPU for data centers and computational facilities



Based on **V2**
(NH)

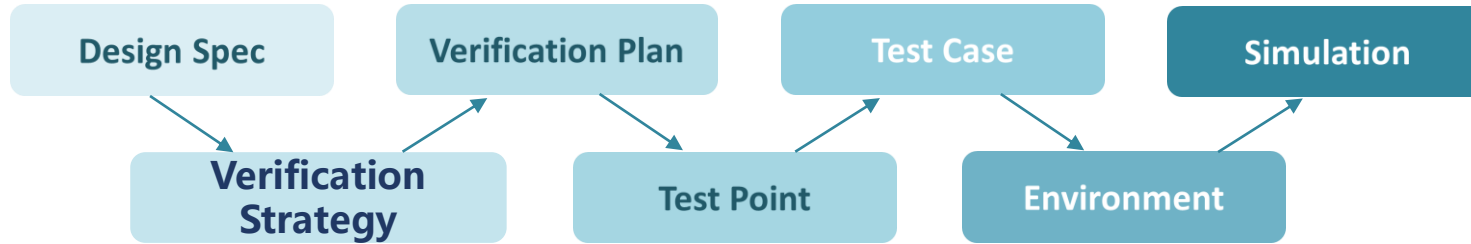
- **Mid Core: Balanced Perf & Efficiency** (v.s. **ARM A76**)

Target Mid-to-High-End General Industry Domain

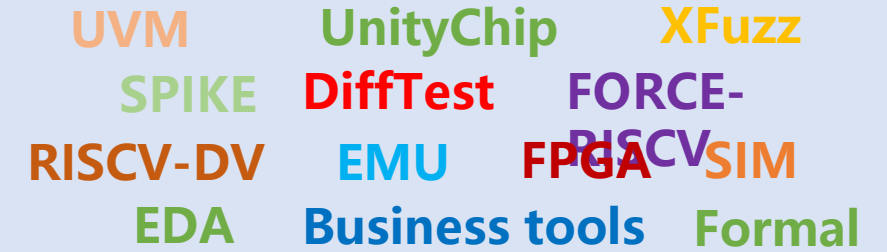
Goal: support broad industrial spectrum including industrial control, automotive, communication, aviation and more

Industrial Level Verification

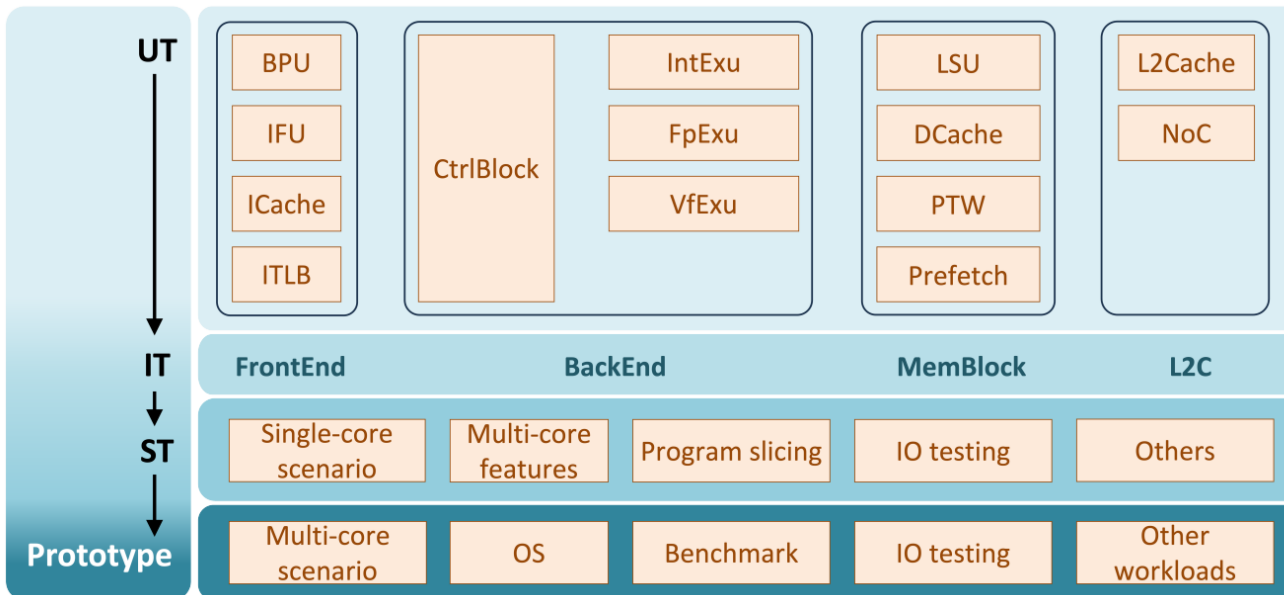
① Process



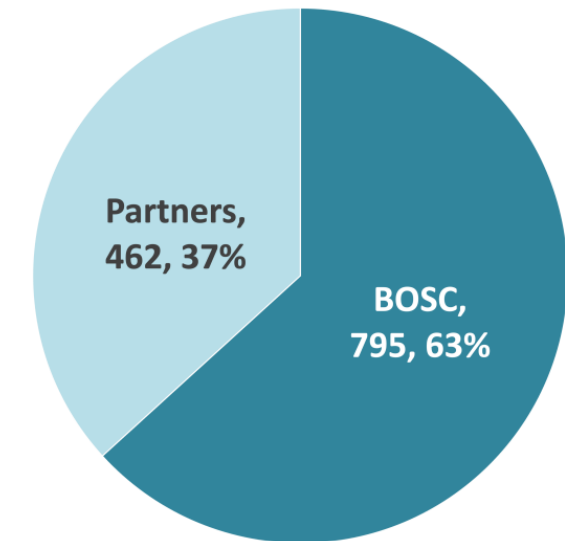
② Tools



③ Verification Hierarchy



④ Partner Collaboration



Source and Number of Reported Bugs



Collaborative Mechanism Based on Open-Source Innovation

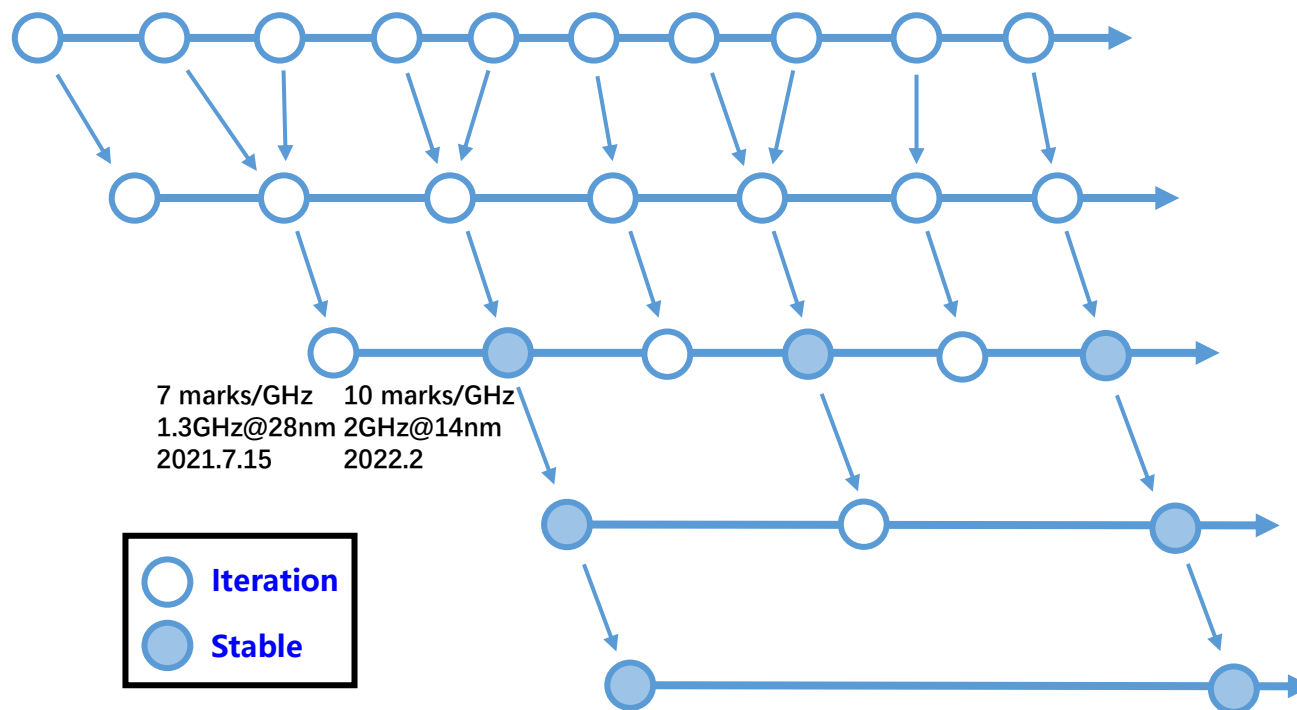
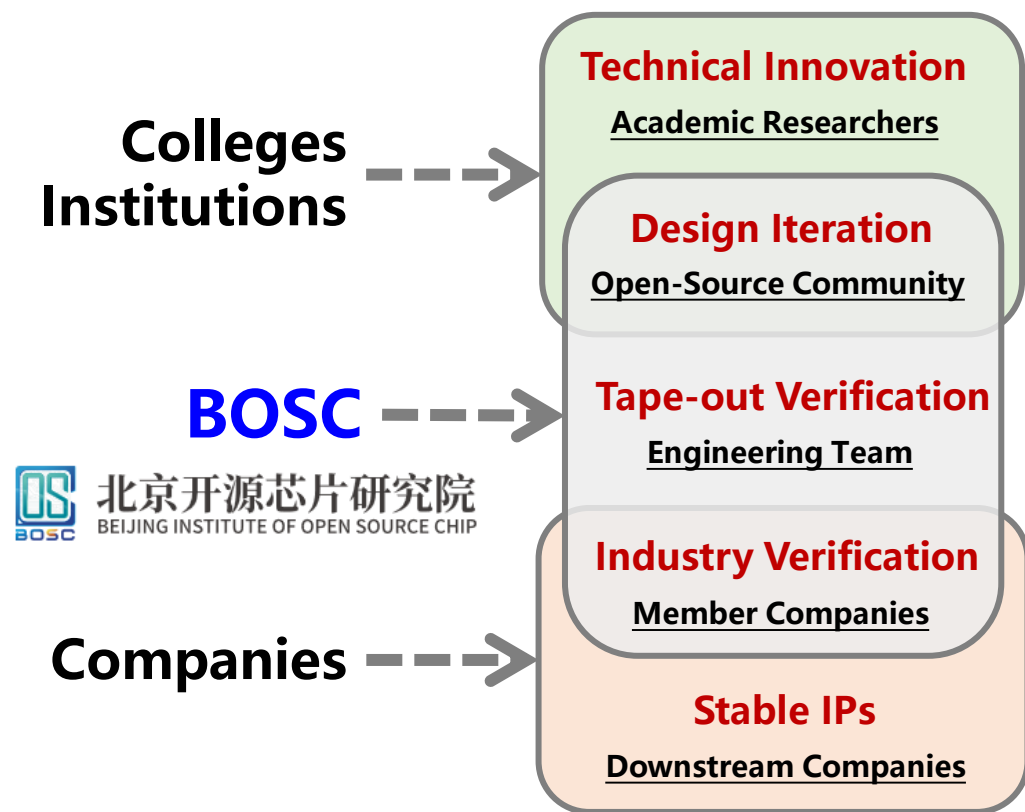
Innovative
Forces



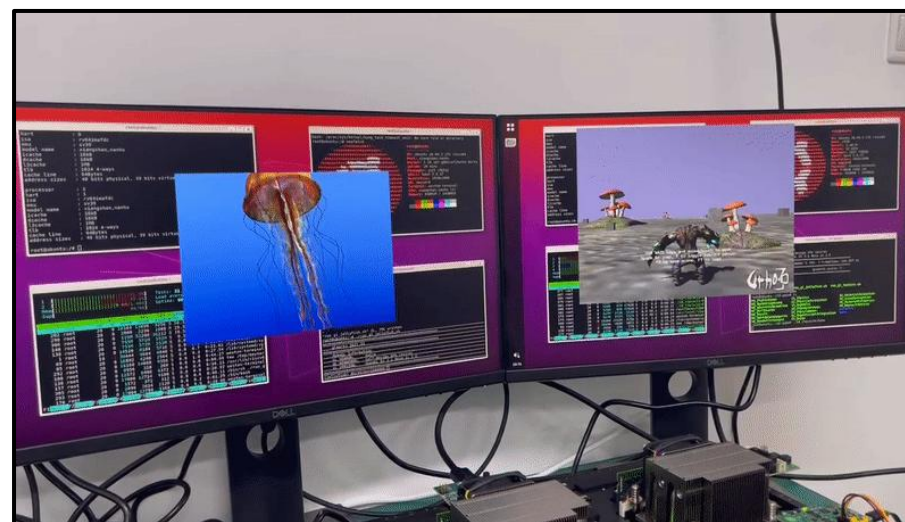
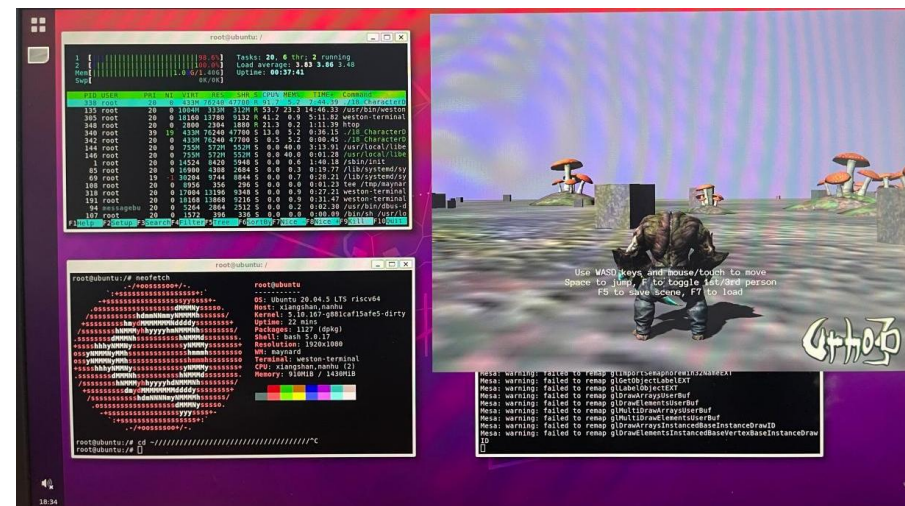
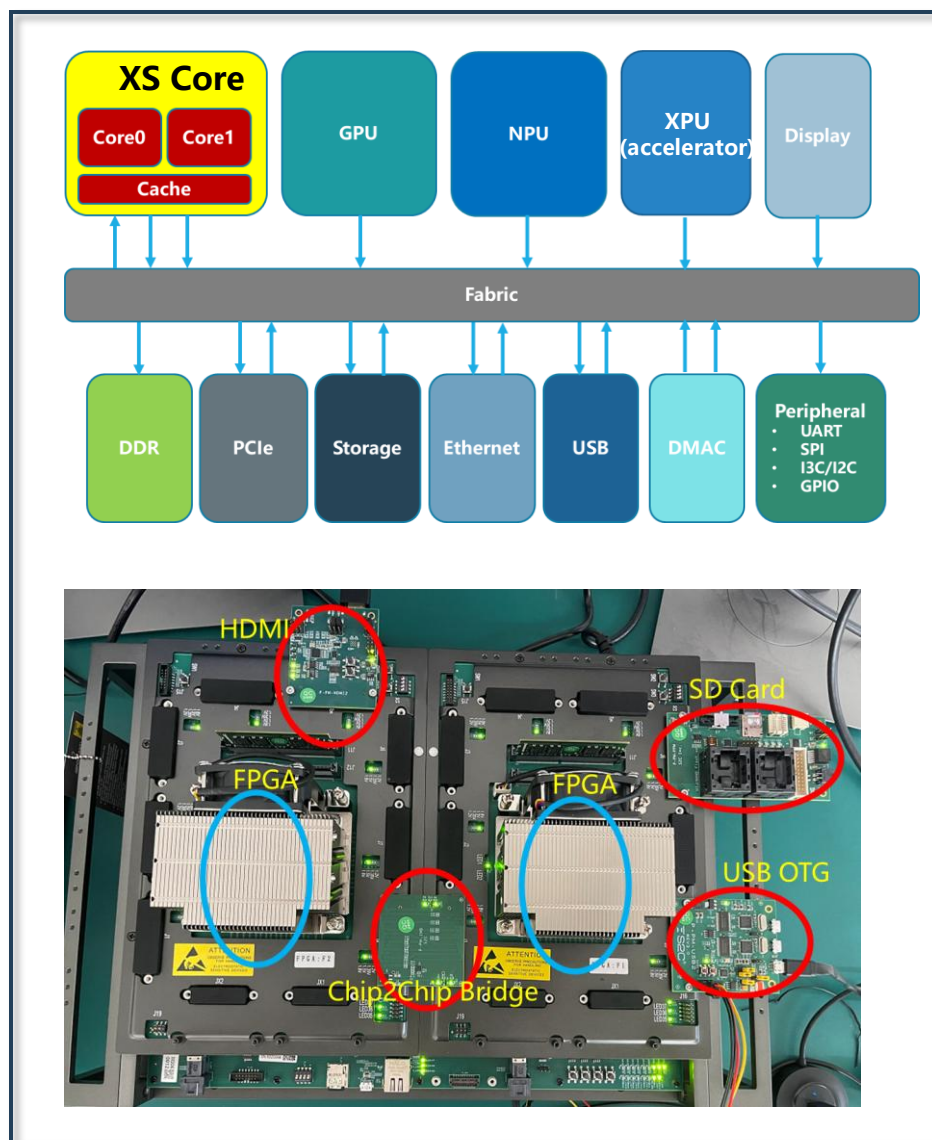
Collaborative
Verification



Tiered Rolling Open-Source Model



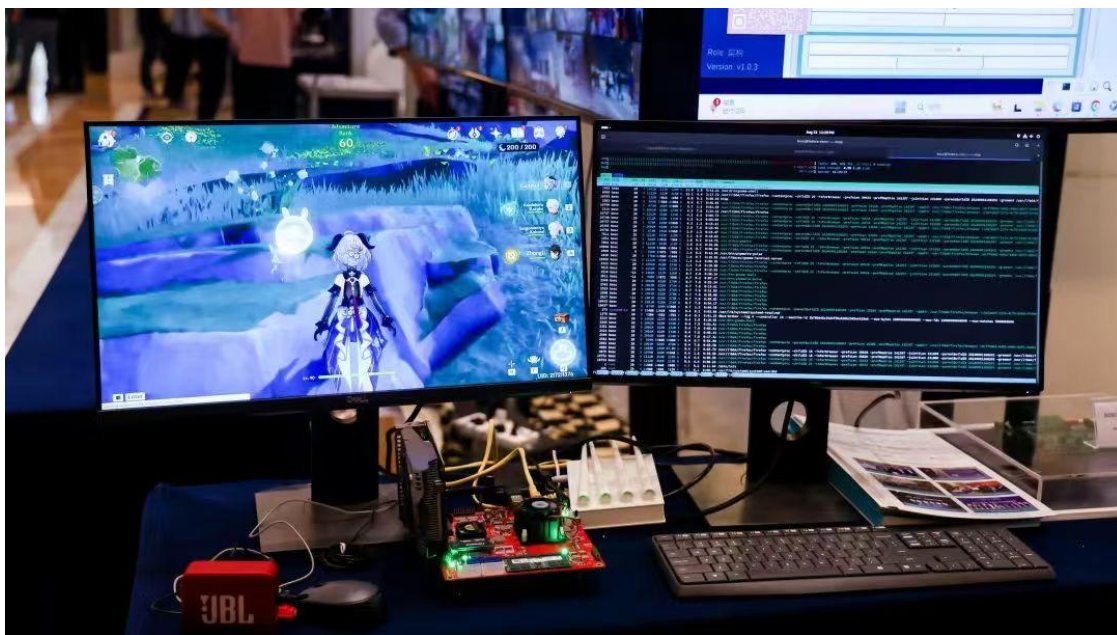
FPGA Prototype in Only Two Weeks



Source: Xinchen Technology

Real Chip of XiangShan Gen 2 Nanhu

- Test chip was back in October 2023
 - Successfully brought up Linux and working with PCIe device (GPU, Ethernet, USB..)



Nanhu test board @ RVSC' 24
Running Desktop Linux & Cloud Game*

The world's first laptop powered by a open-source RISC-V processor

ISCAS
中国科学院软件研究所
Institute of Software Chinese Academy of Sciences

Ruyi Book	
CPU	"XiangShan Nanhu" (RV64GCBK), up to 2.5GHz
Memory	8GB DDR5 4800MT/s
GPU	AMD RX 550
USB	2x USB3
Ethernet	2x 2.5Gbps Ethernet Port
Display	1x 14-inch LCD Display 1x HDMI, up to 4K
TouchPad	Support 9 kinds of gesture operation
Audio	Built-in high-quality speakers.
Dimensions	315*233*25mm

Powered by

XIANGSHAN
"XiangShan" high-performance open-source RISC-V processor

Ruyi Book

inchi 英趣智能 milkV

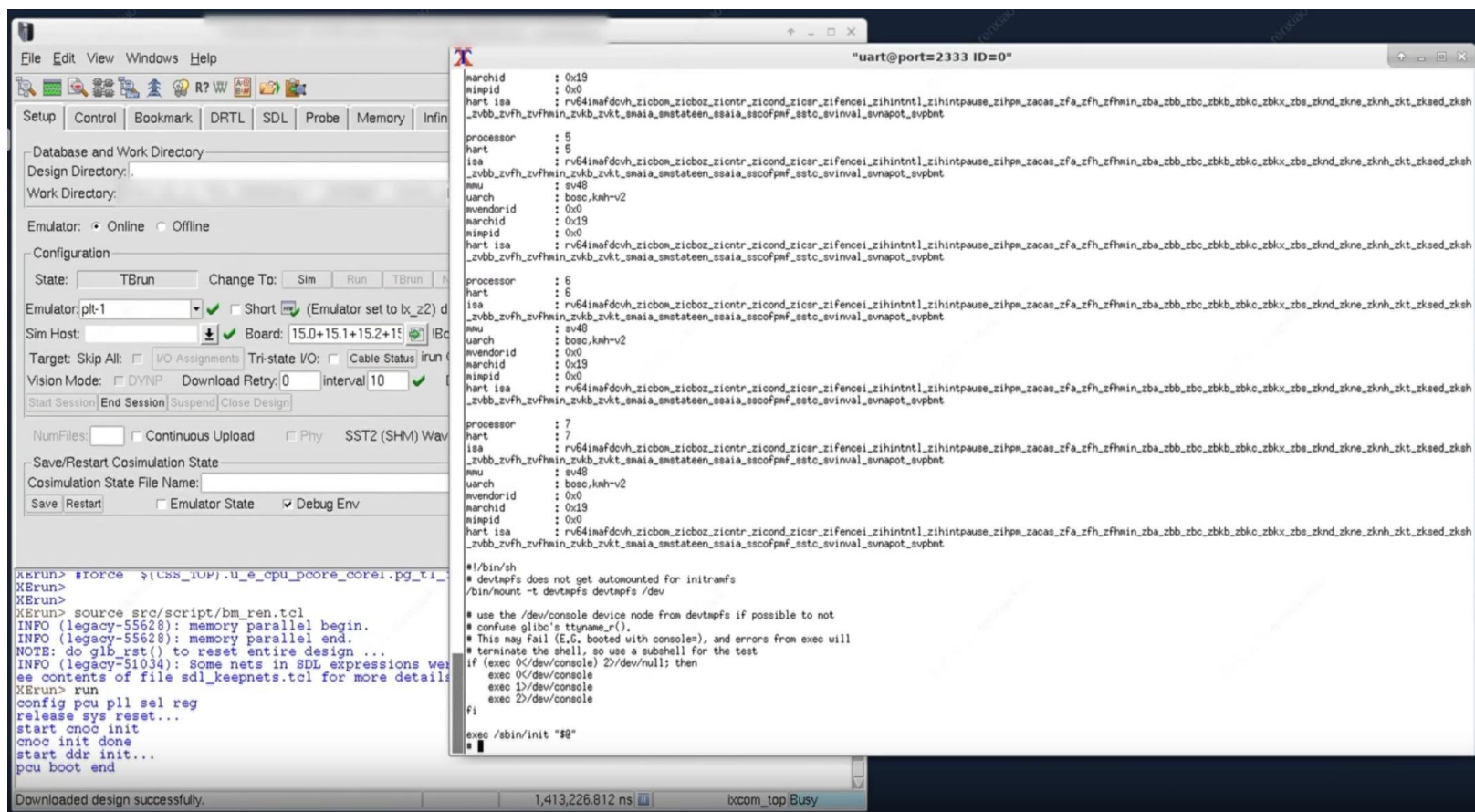
Ruyi XiangShan Book
Design by ISCAS, Inchi, MilkV

* Genshin Impact· Cloud, only ~ 1 fps, just for fun

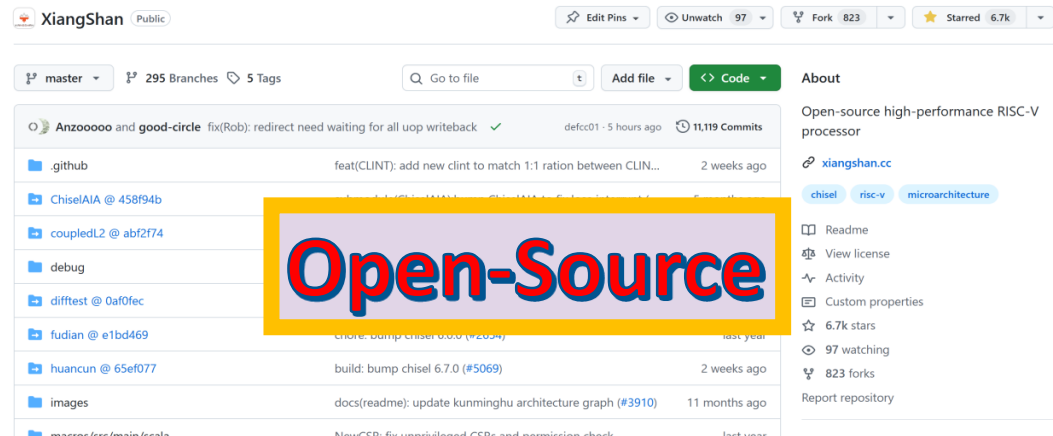


Linux Boot on 8-Core XiangShan Gen 3 Kunminghu V2

- Lanxin Computing has successfully launched Linux on an 8-core SoC built on the Kunminghu V2

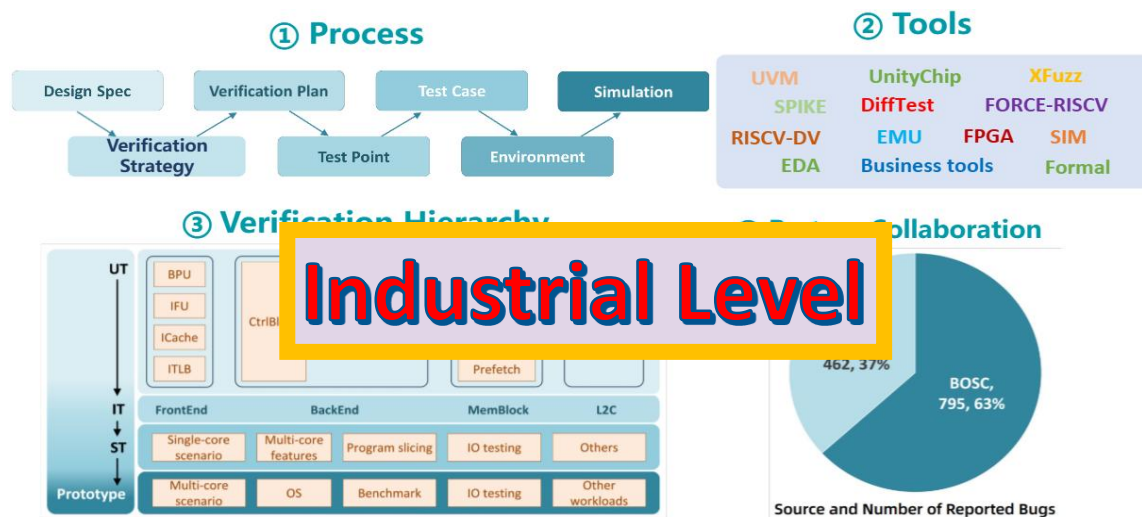
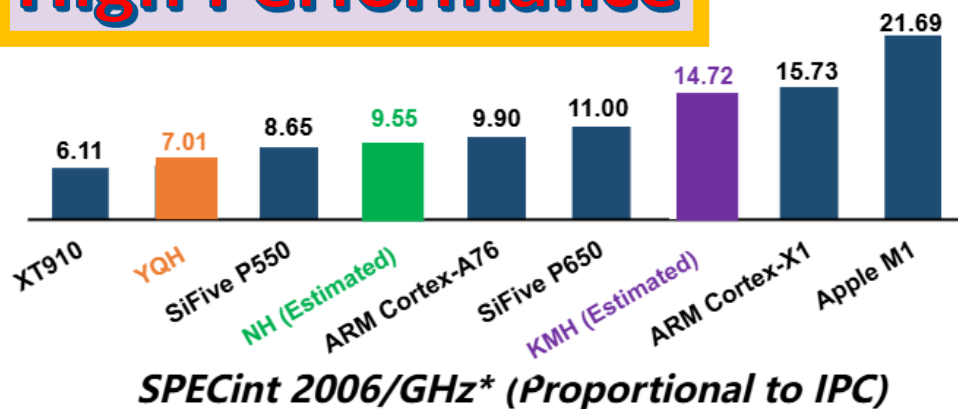


XiangShan: A RISC-V Processor



> 6.7K stars, > 820 forks on GitHub

High Performance



User-Friendly Infrastructure



First Step to Agile Design: Use Chisel

• 2018: quantitative experiments between Chisel and Verilog

- **Task #1:** Design an L2 Cache for RISC-V Rocket-chip core
- **Who:** A 5-year engineer vs. a senior student

	A 5-year Engineer	An Undergraduate
Experience	Familiar w/ OpenSparc T1; Modified Xilinx Cache	Course projects on CPU design; Using Chisel for 9 months
Language	Verilog	Chisel
Time	6 weeks	3 days
LOCs	~1700	~350
Results	Unable to boot Linux	Boot Linux on multicore RISC-V; support NIC w/ DMA mode

- **1st Round results:** Chisel is more productive than Verilog by **14X** with only **1/5** LOC

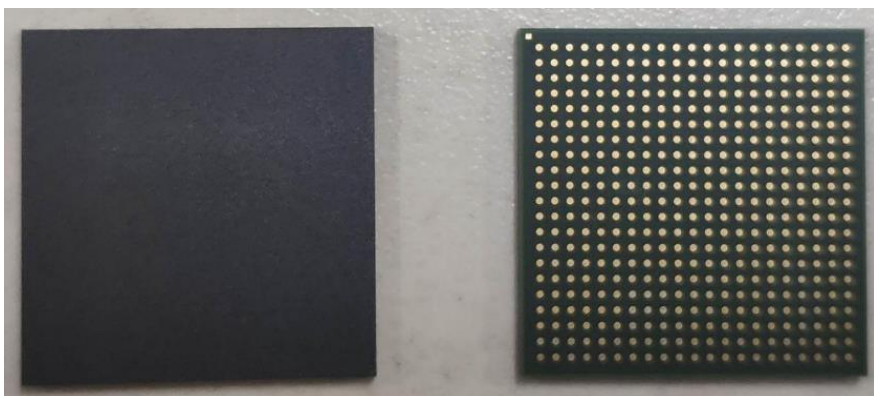
- **Task #2:** Translate the Verilog codes into Chisel
- Evaluated on FPGA (xc7v2000tfhg1716-1), Vivado 2017.01
- **Who:** A junior student who never knew Chisel

	Verilog	Chisel (direct translation)	Chisel-opt (adv. features & libs)
Freq./MHz	135.814	136.388 (+0.42%)	154.107 (+13.47%)
Power/W	0.770	0.749 (-2.73%)	0.749 (-2.73%)
LUT Logic	5676	6422 (+13.14%)	2594 (-54.30%)
LUT Storage	1796	1264 (-29.62%)	1492 (-16.93%)
FF	4266	3638 (-14.72%)	747 (-82.49%)
LOCS	618	470 (-23.95%)	155 (-74.92%)

- **2nd Round results:** Chisel can achieve better PPA than verilog

Yu Zihao, Liu Zhigang, Li Yiwei, Huang Bowen, Wang Sa, Sun Ninghui, Bao Yungang. Practice of Chip Agile Development: Labeled RISC-V. Journal of Computer Research and Development, 2019, 56(1): 35-48.

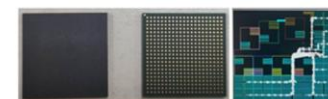
• 2020: 28-nm tape-out of an 8-core labeled RISC-V processor



16-node prototype case



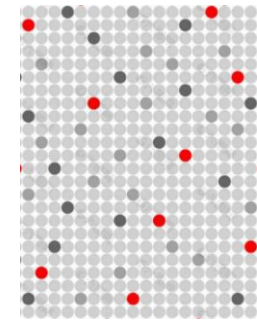
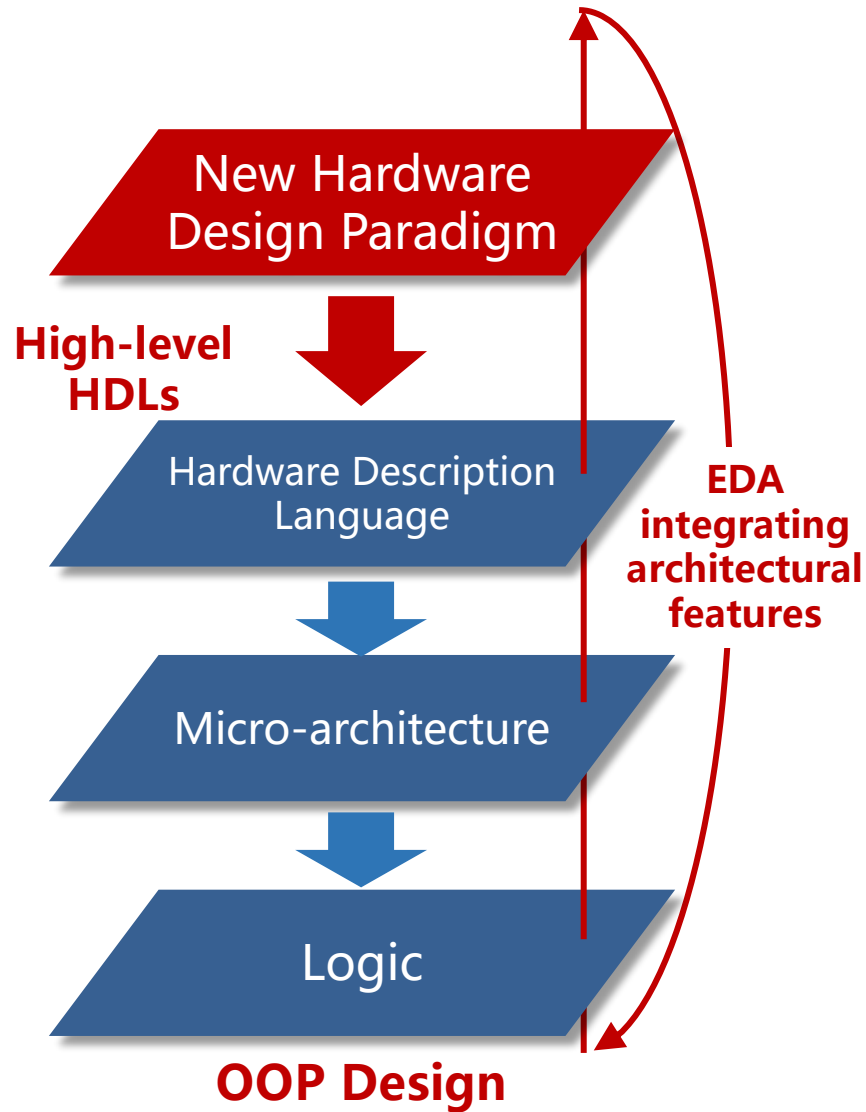
single node board



"BeiHai 100"
Labeled RISC-V Chip

- RV64GC指令集
- 单发射顺序9级流水线
- 内置标签化冯诺依曼结构技术
- 8核/2MB L2 Cache
- ChipLink前端总线
- **1.2GHz@ 28nm**
- Wafer out/WB BGA封装
- 最大支持32GB DDR4内存
- 2*千兆以太网
- 1*PCIe3.0 RC x4

New HDL → New Design Paradigm

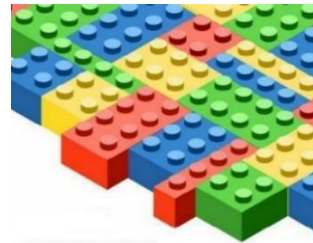


Pixel
(gate-level)

Circuit-oriented
Processor
Design



Process
or



Object
(Reusable Modules)

Object-
oriented
Processor
Design



Process
or



Configurable





What's Missing in Agile Hardware Design? Verification!

What's Missing in Agile Hardware Design? Verification!

Babak Falsafi, *Fellow, ACM, IEEE*

Parallel Systems Architecture Laboratory, Institute of Computer and Communication Sciences, School of Computer and Communication Sciences, Ecole polytechnique fédérale de Lausanne CH-1015 Lausanne

E-mail: babak.falsafi@epfl.ch

Agile hardware design is an approach to developing hardware systems that draws inspiration from the principles and practices of agile software development. It emphasizes collaboration, flexibility, iterative development, and quick adaptation to changing requirements. In agile hardware design, the focus is on delivering functional hardware systems in shorter development cycles while maintaining high-quality and customer satisfaction.

In particular, agile hardware design is of great interest in the open-source hardware community. Open-source hardware development —such as RISC-V— is at the forefront of initiatives to democratize hardware and drive innovation in chip design forward. Agile design is instrumental for the RISC-V community because it supports rapid iteration, accommodates the evolving RISC-V standard and the addition of custom extensions, improves community collaboration and time-to-market, and addresses the design challenges associated with complex architectural features.

Among significant innovations based on agile hardware design is the recently announced XIANGSHAN RISC-V core which is currently the highest performing RISC-V out-of-order microprocessor core with single-thread performance exceeding both existing RISC-V cores and a state-of-the-art ARM core, Cortex-A76. The creators of this platform have published their agile design methodology in a flagship computer architecture venue, MICRO, with a paper that has been selected through peer review to be among the best dozen papers in all of computer architecture in one year for publication in IEEE Micro Top Picks.

A key contributor to this breakthrough has been integrating hardware verification into the agile methodology. Hardware verification is crucial in designing digital platforms, as it ensures that semiconductor chips operate correctly and reliably according to the architecture specifications. Verification guarantees compliance with standards, and helps detect and rectify design errors, validate system-level functionality, optimize performance and power consumption, and enhance hardware reliability and safety. It plays a fundamental role in creating robust and dependable CPUs that meet the requirements of various applications and workloads.

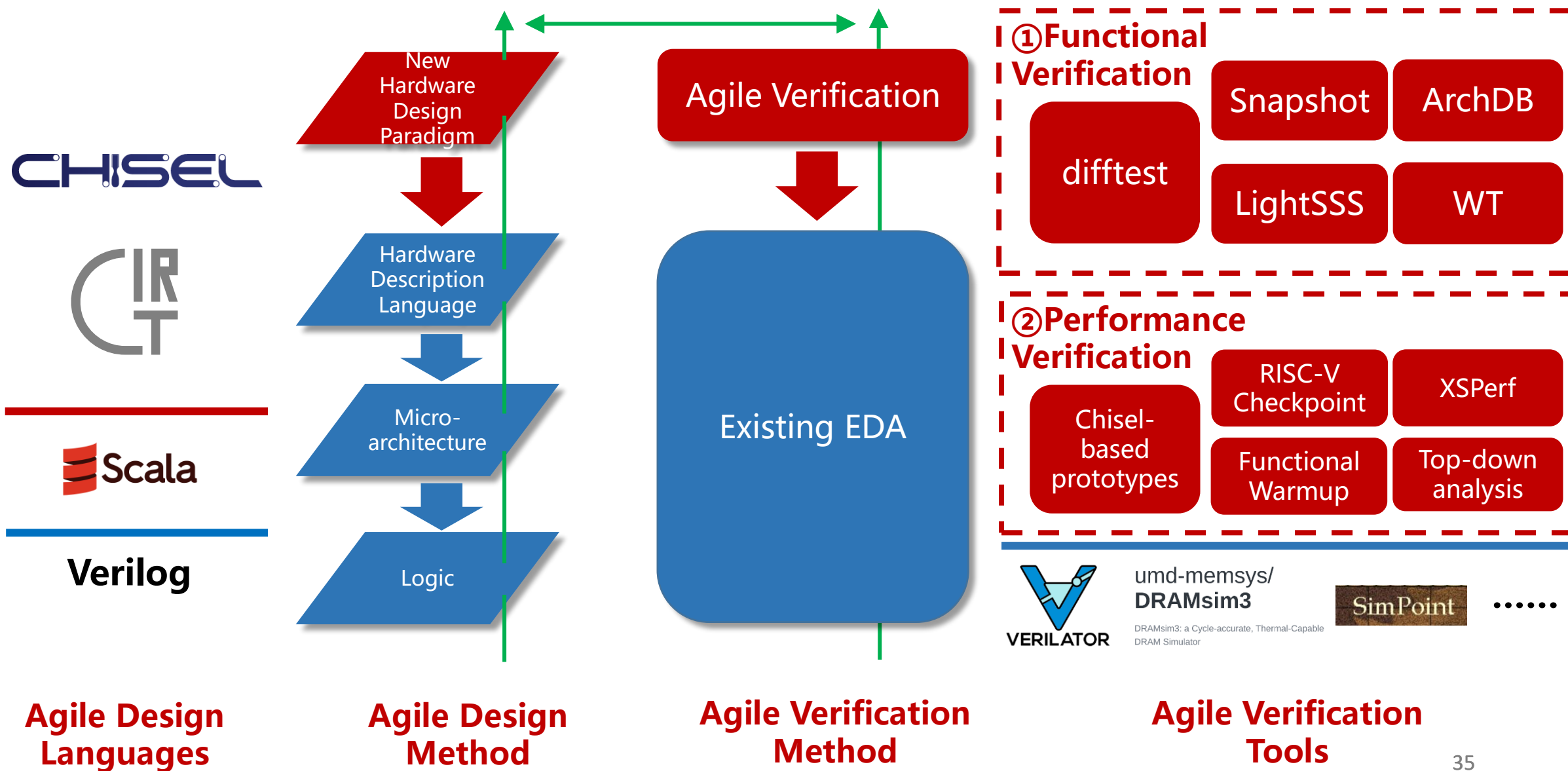


Babak Falsafi

Professor, EPFL

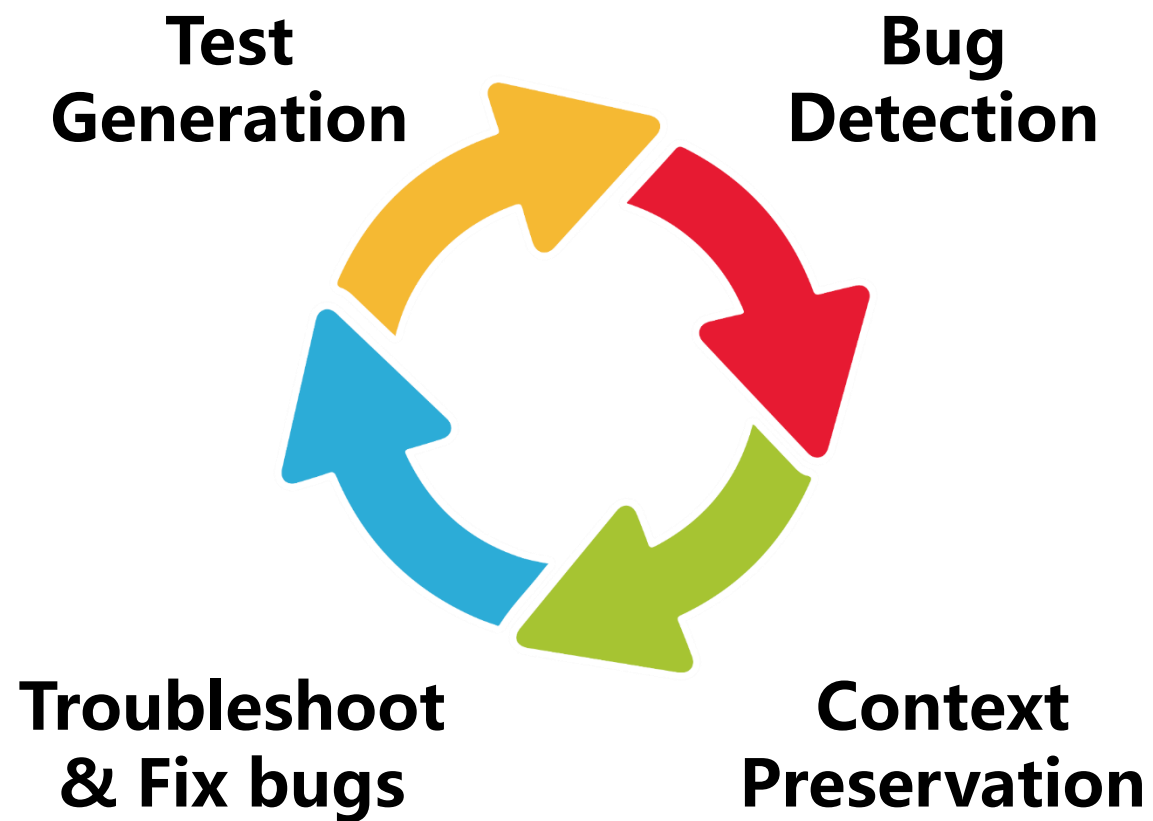
ACM/IEEE Fellow

Agile Verification is Challenging

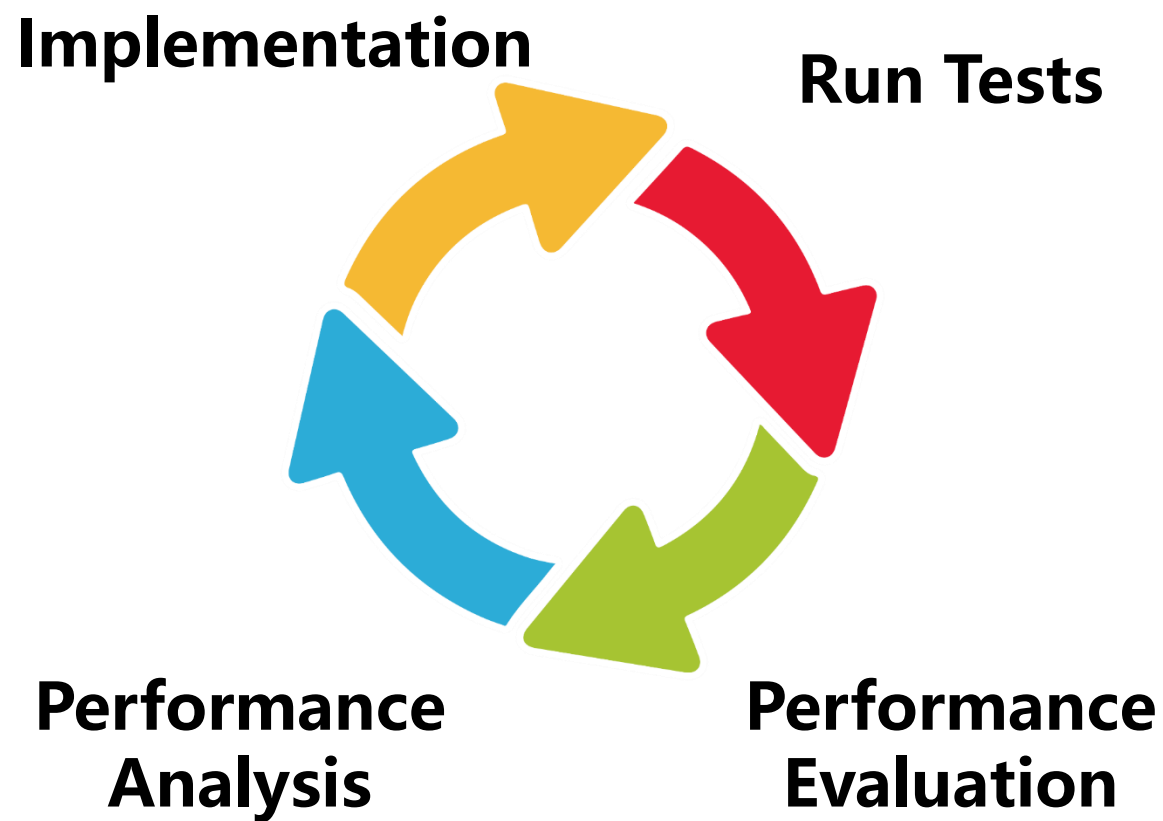


MinJie Loop

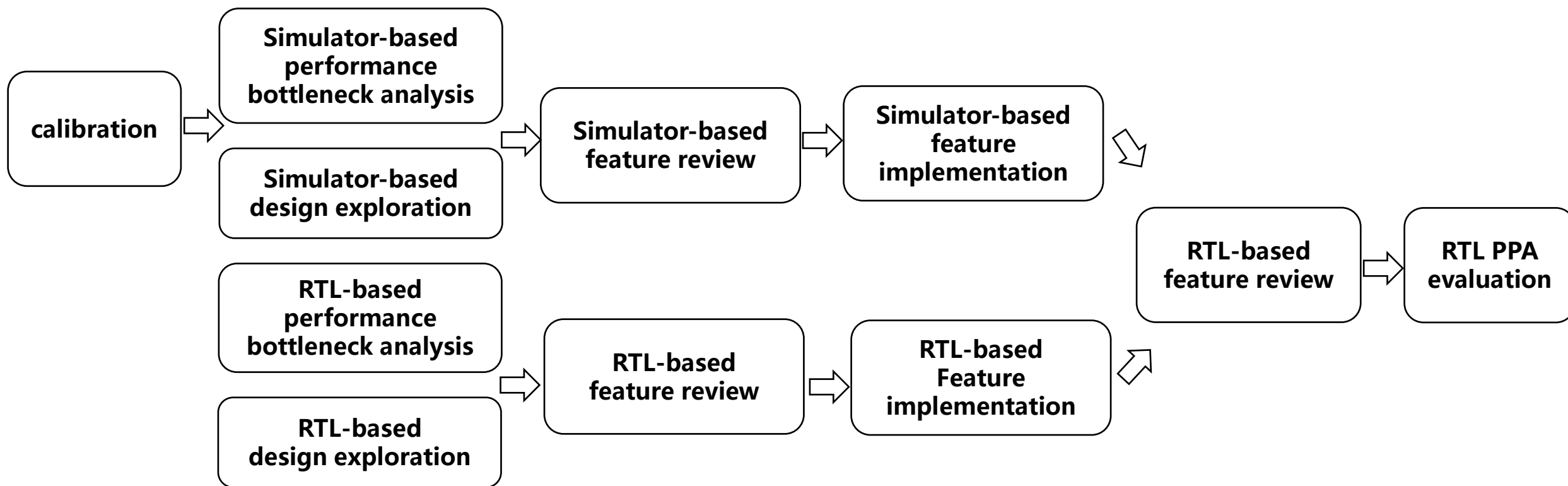
Functional Verification Loop



Performance Optimization Loop



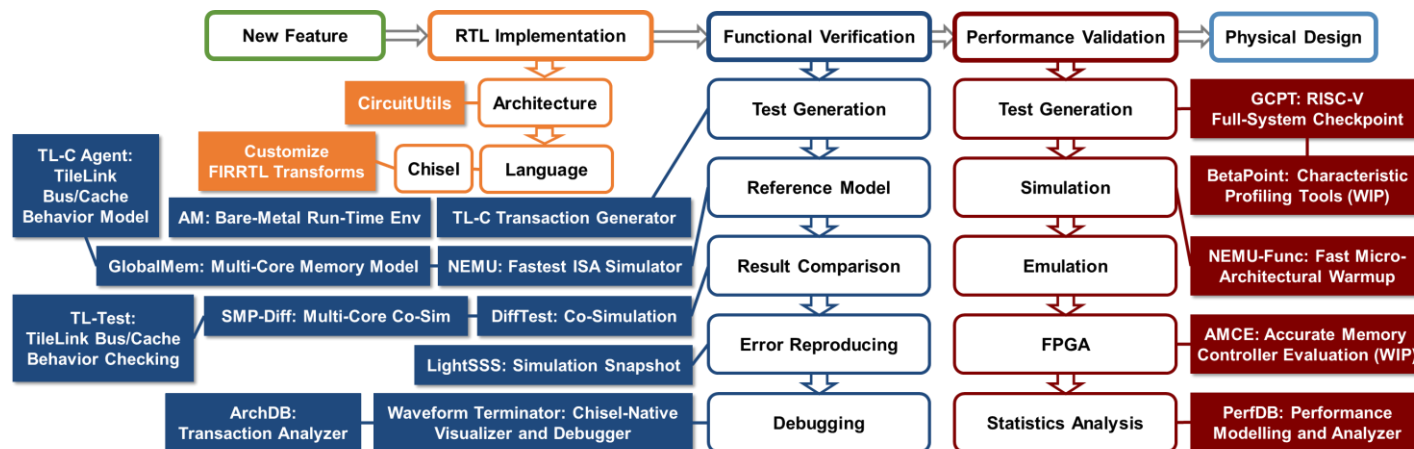
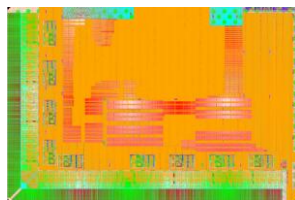
XS-Gem5: Calibrated Gem5 Simulator





MinJie: Open & Agile Verification Toolchain

- Infrastructure is the key outcome of the XiangShan Project
- Open source to benefit both academia and industry





Advanced Agile Chip Design Methodology

- New method and toolchain for agile chip design
- Developed more than 17 new tools to solve agile verification problems
- **MICRO 2022 → IEEE Micro Top Picks**

2022 55th IEEE/ACM International Symposium on Microarchitecture (MICRO)



Towards Developing High Performance RISC-V Processors Using Agile Methodology

Yinan Xu[†], Zihao Yu^{*}, Dan Tang^{*‡}, Guokai Chen^{*†}, Lu Chen^{*†}, Lingrui Gou^{*†}, Yue Jin^{*†}, Qianruo Li^{*†}, Xin Li^{*†}, Zuojun Li^{*†}, Jiawei Lin^{*†}, Tong Liu^{*}, Zhigang Liu^{*}, Jiazhan Tan^{*}, Huaqiang Wang^{*†}, Huizhe Wang^{*†}, Kaifan Wang^{*†}, Chuanqi Zhang^{*†}, Fawang Zhang^{||}, Linjuan Zhang^{*†}, Zifei Zhang^{*†}, Yangyang Zhao^{*}, Yaoyang Zhou^{*†}, Yike Zhou^{*}, Jiangrui Zou^{||}, Ye Cai^{||}, Dandan Huan[¶], Zusong Li[¶], Jiye Zhao[¶], Zihao Chen[§], Wei He[§], Qiyan Quan[§], Xingwu Liu^{**}, Sa Wang^{*†}, Kan Shi^{*}, Ninghui Sun^{*†} and Yungang Bao^{*†}

^{*}State Key Lab of Processors, Institute of Computing Technology, Chinese Academy of Sciences, China

[†]University of Chinese Academy of Sciences, China

[‡]Beijing Institute of Open Source Chip, China

[§]Peng Cheng Laboratory, China

[¶]Beijing VCore Technology Co., Ltd., China

^{||}Shenzhen University, China

^{**}Dalian University of Technology, China

THEME ARTICLE: TOP PICKS FROM THE 2022 COMPUTER ARCHITECTURE CONFERENCES

Toward Developing High-Performance RISC-V Processors Using Agile Methodology

Yinan Xu^{IC} and Zihao Yu^{IC}, State Key Lab of Processors, Institute of Computing Technology, Chinese Academy of Sciences, Beijing, 100190, China

Dan Tang^{IC}, Beijing Institute of Open Source Chip, Beijing, 100080, China

Ye Cai^{IC}, Shenzhen University, Shenzhen, 518060, China

Dandan Huan^{IC}, Beijing VCore Technology, Beijing, 100190, China

Wei He^{IC}, Peng Cheng Laboratory, Shenzhen, 518060, China

Ninghui Sun^{IC} and Yungang Bao^{IC}, State Key Lab of Processors, Institute of Computing Technology, Chinese Academy of Sciences, Beijing, 100190, China



DiffTest-H: Hardware-Accelerated Co-Simulation Verification

- **13.8 MHz** on FPGA, instruction-level debugging
- Deployed on **XiangShan**, with **151 bugs uncovered**
- **Open-Source** at [github/OpenXiangShan/difftest](https://github.com/OpenXiangShan/difftest)

2:30 PM, Tuesday, Oct. 21
Room C, Session 7C:
Reliability, Fault-tolerance



中国科学院计算技术研究所
INSTITUTE OF COMPUTING TECHNOLOGY, CHINESE ACADEMY OF SCIENCES



北京开源芯片研究院
BEIJING INSTITUTE OF OPEN SOURCE CHIP



香山开源处理器社区
XiangShan Open-Source Processor Community

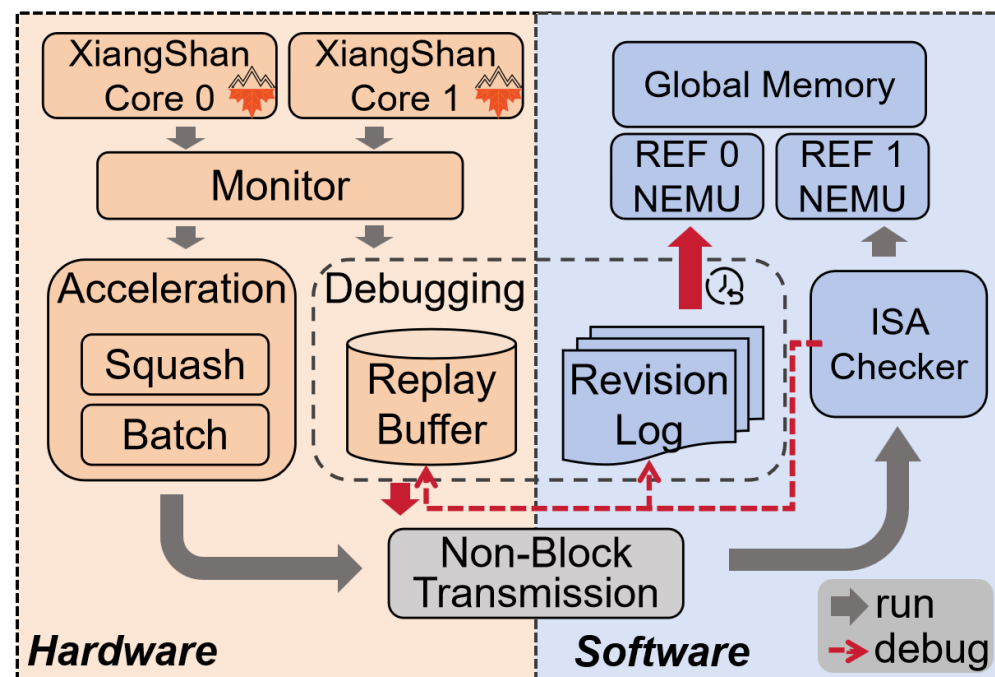
DiffTest-H: Toward Semantic-Aware Communication in Hardware-Accelerated Processor Verification

Kunlin You^{1,2}, Yinan Xu¹, Kehan Feng³, Luoshan Cai^{1,2}, Yaoyang Zhou³ and Yungang Bao^{1,2}

¹ State Key Lab of Processors, Institute of Computing Technology, Chinese Academy of Sciences

² University of Chinese Academy of Sciences

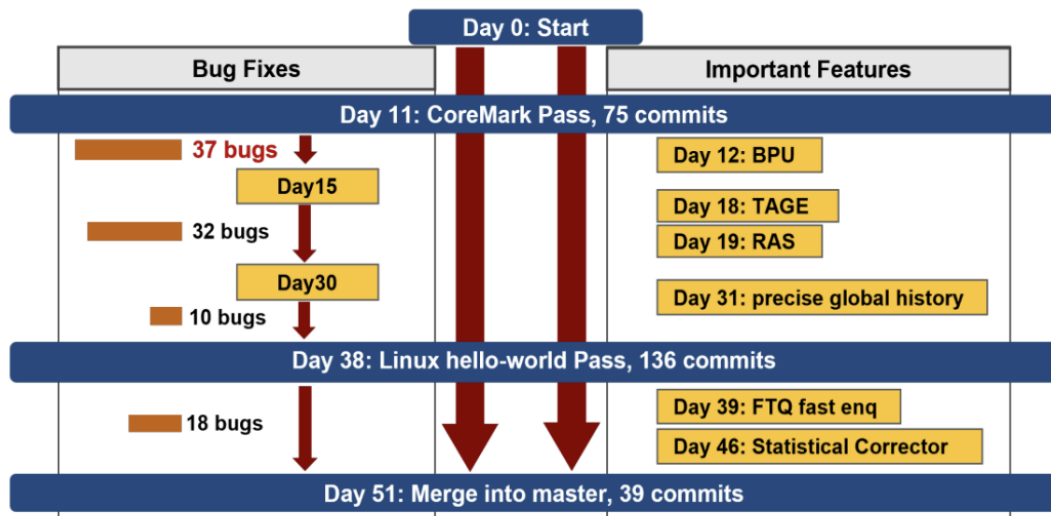
³ Beijing Institute of Open Source Chip





An Effective Infrastructure for Research

① Micro-architecture Optimization



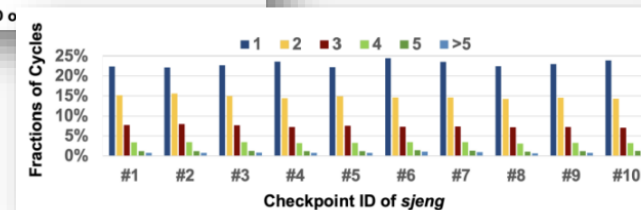
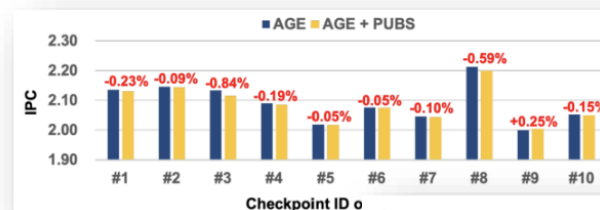
- 3 graduate students
- 11 days for a functionally correct prototype
- 37 bugs in 5 days
- 38 days to boot Linux with BPU
- 51 days for the overall frontend architecture

② Paper Reproduction

2018 51st Annual IEEE/ACM International Symposium on Microarchitecture

Performance Improvement by Prioritizing the Issue of the Instructions in Unconfident Branch Slices

One third-year PhD student, 200 minutes on XiangShan





-
- The image shows a laptop on the left with the OpenSBI boot menu displayed on its screen. The menu lists various boot options and system information. A red arrow points from the laptop screen to a custom-built PC on the right. The PC case is open, showing internal components including a motherboard, RAM, and a fan. A red box at the bottom contains terminal output from the OpenSBI boot process.
- ```

3.295436] 9pnet: installing 9p2000 support
3.299728] Key type dns_resolver registered
4.257062] Freeing unused kernel image (initmem) memory: 38212K
4.262726] Run /init as init process

#
~#
cat /proc/
1/ 26/ 7/ interrupts partitions
10/ 27/ 8/ iomem scsi/

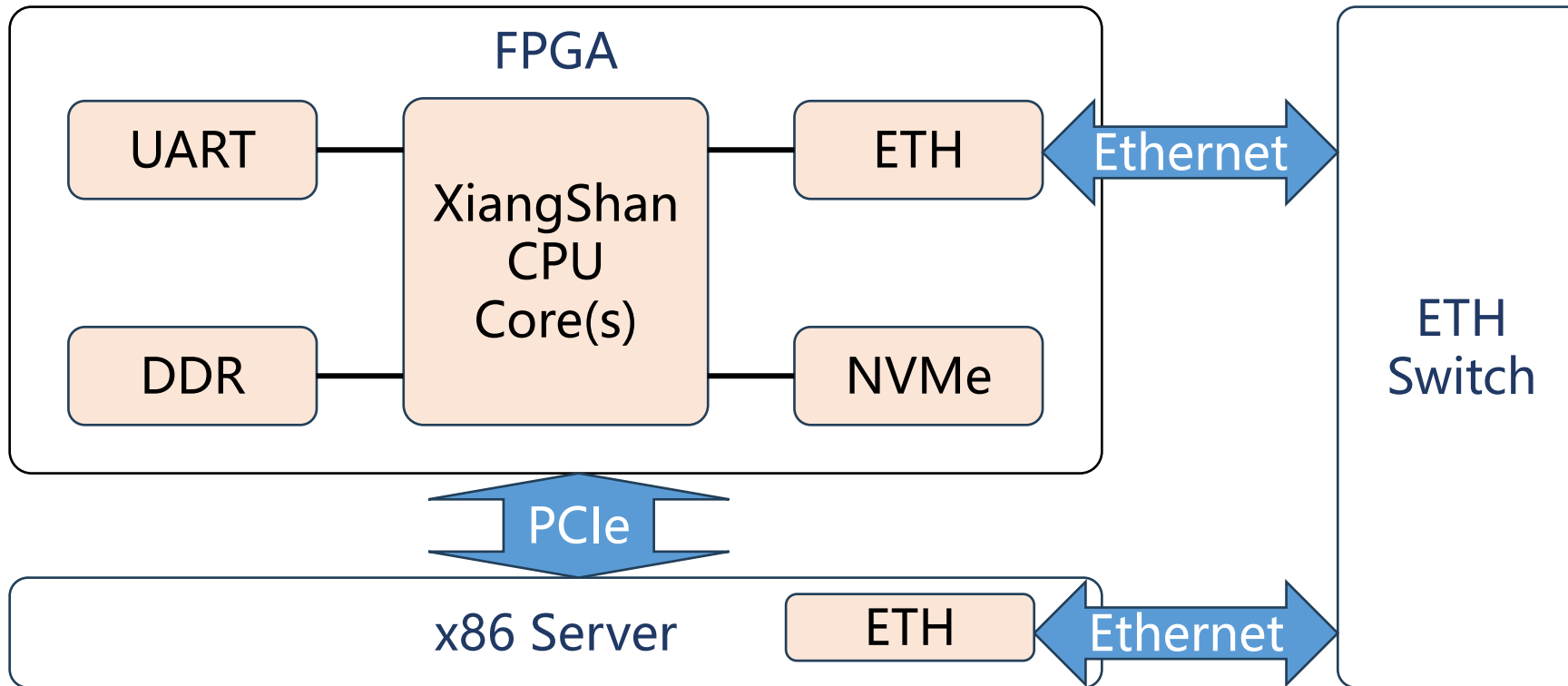
```



## Start Linux Locally with NEXST



# NEXST: Next Environment for XiangShan Target



- Various peripherals
  - UART, Ethernet (MAC & Phy), M.2 NVMe, and Other PCIe peripherals



# Current FPGA Board Support: VU37P/VU19P

| FPGA Chip     |                         | Xilinx VU37P (2 public, 1 private)                                                  |                                                                                       |                                                                                       | Xilinx VU19P (1 public + 1 private)                                                   |                                                                                       |
|---------------|-------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| FPGA Board    |                         | Xilinx VCU128                                                                       | Xilinx U280                                                                           | ICT NM37                                                                              | Corigine MimicTurbo GT                                                                | ICT NP19A                                                                             |
| Board Config. | M.2 Interface           | ×                                                                                   | ×                                                                                     | ✓ (1x M.2/PCIe)                                                                       | ×                                                                                     | ✓ (4x M.2/PCIe)                                                                       |
|               | RJ45 Interface          | ✓                                                                                   | ×                                                                                     | ✓                                                                                     | ✓                                                                                     | ✓                                                                                     |
|               | PCIe Card Format        | Non-standard                                                                        | ✓                                                                                     | ✓                                                                                     | ✓                                                                                     | ✓                                                                                     |
|               | DDR4                    | 4.5GB                                                                               | 32GB (2x DIMM)                                                                        | 32GB (2x SO-DIMM)                                                                     | 16GB                                                                                  | 32GB (2x SO-DIMM)                                                                     |
| CPU Config.   | Octa-Core Rocket-H ext. | To Be Tested                                                                        | To Be Tested                                                                          | ✓                                                                                     | ✓                                                                                     | ✓                                                                                     |
|               | Single-Core Nanhu-G     | ✓                                                                                   | ✓                                                                                     | ✓                                                                                     | ✓                                                                                     | ✓                                                                                     |
|               | Dual-Core Nanhu-G       | ×                                                                                   | ×                                                                                     | ×                                                                                     | ✓                                                                                     | ✓                                                                                     |
|               | Single-Core Kunminghu   |  |  |  |  |  |
| Board Photo   |                         | /                                                                                   |                                                                                       |                                                                                       |                                                                                       |                                                                                       |



# An Open-Source FPGA Environment for XiangShan

**1:00 PM, TODAY  
Same Room HERE**

- **FPGA SoC Environment for XiangShan**
  - Various peripherals support
  - Multiple FPGA boards support
- CI/CD for XiangShan Development
- Run XiangShan locally or in SERVE Cloud
- **Open-Source: <https://github.com/NEXST-TOOL/nexst.git>**
- ***MICRO Tutorials this afternoon at the same place!***
  - REMU: An FPGA-based Open-source Replayable EMUlator for System-level Evaluating and Debugging Processors



# An Effective Infrastructure for Research

- **Topic: Computer Architecture**

- XiangShan: a realistic out-of-order RISC-V implementation with industry-competitive performance and an active open-source community
- MinJie provides the toolchains
- *Microarchitecture, accelerators, novel architectures, profiling, systems, benchmarking, security, compilers, ...*



midgard

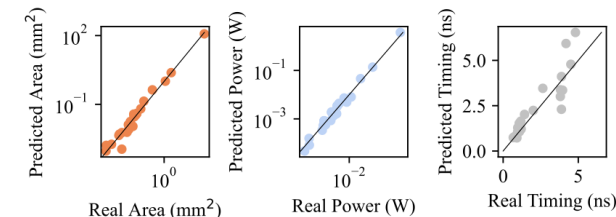
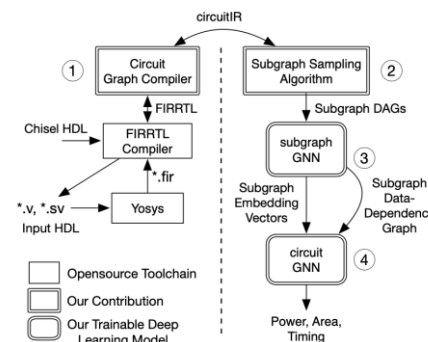
EPFL

Imprecise Store Exceptions, ISCA'23

Single Address Space Faas with Jord, ISCA'25

- **Topic: Agile Chip Development**

- XiangShan is a progressive, configurable, complicated, challenging benchmark
- MinJie provides a good startpoint
- *HDLs, verification, performance, power, area, prototyping, DFT, synthesis, placement, routing,*



SNS v2, MICRO'23 (Duke University)



# XiangShan achieves L2.5

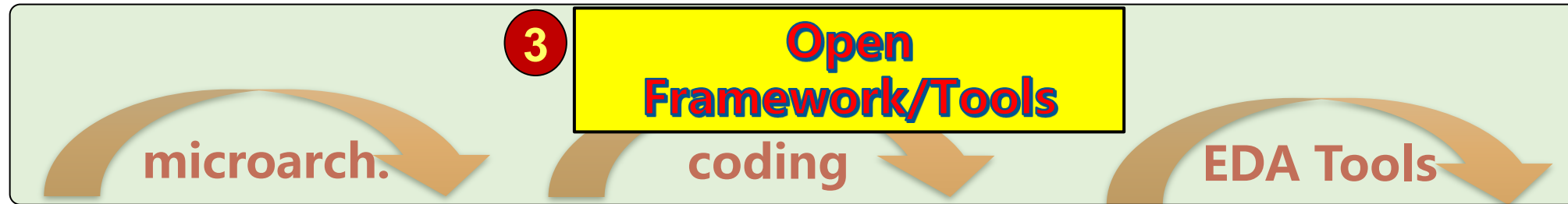
L1: OPEN ISA



L2: OPEN Design/Implementation



L3: OPEN Framework/Tools

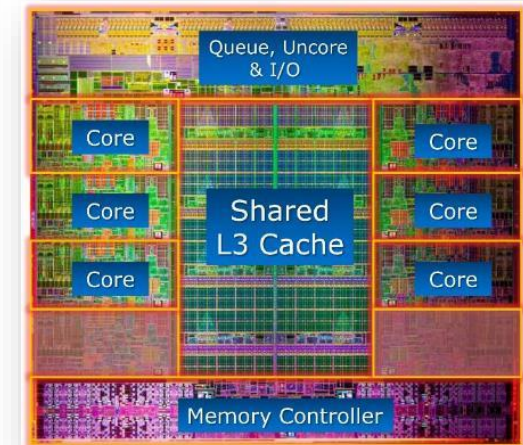


ISA Spec.

Docs

RTL

Layout



**1 Open ISA**

**2 Open Design/Implt**

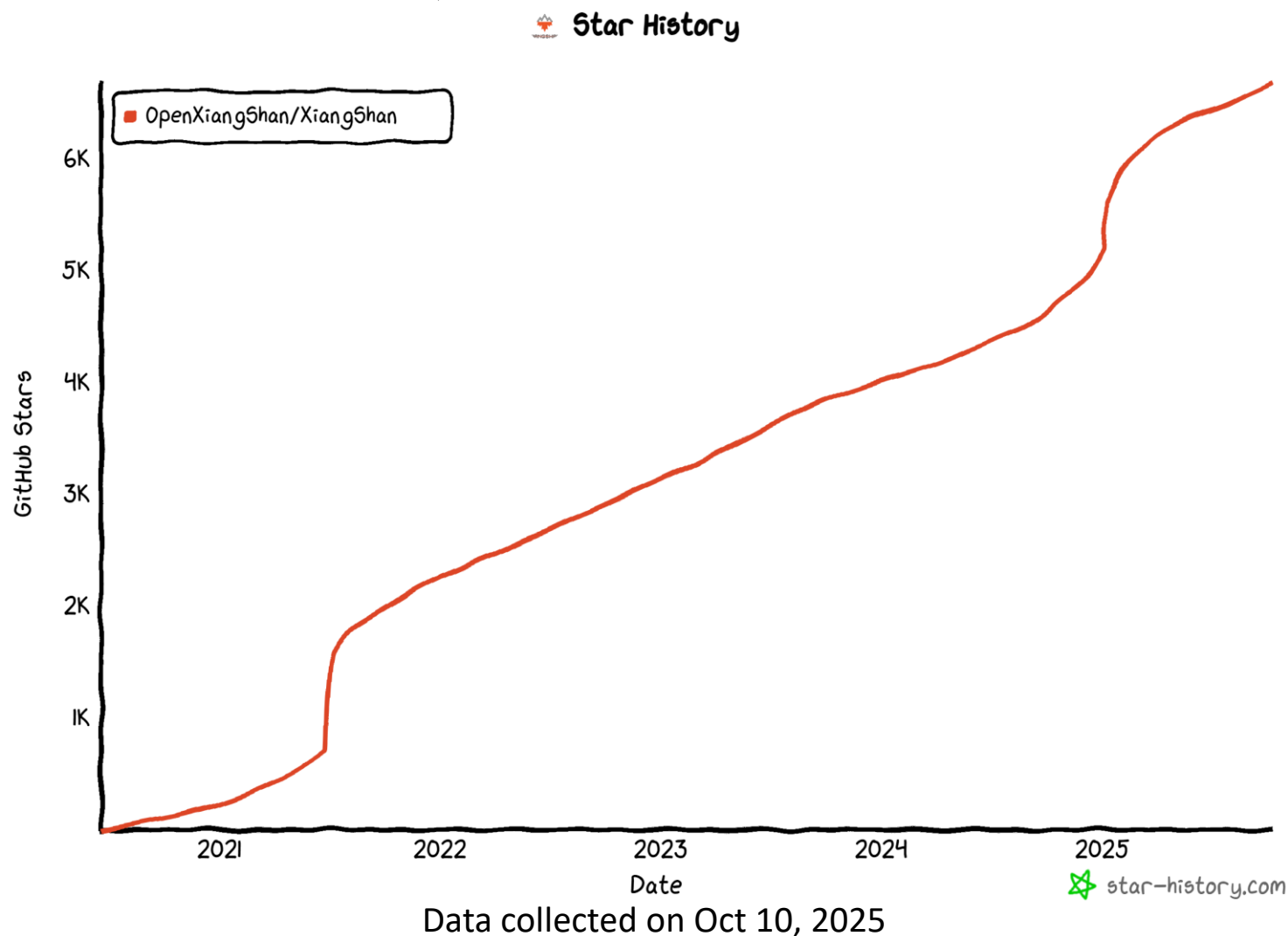
# **Prosperous Open-Source Community**

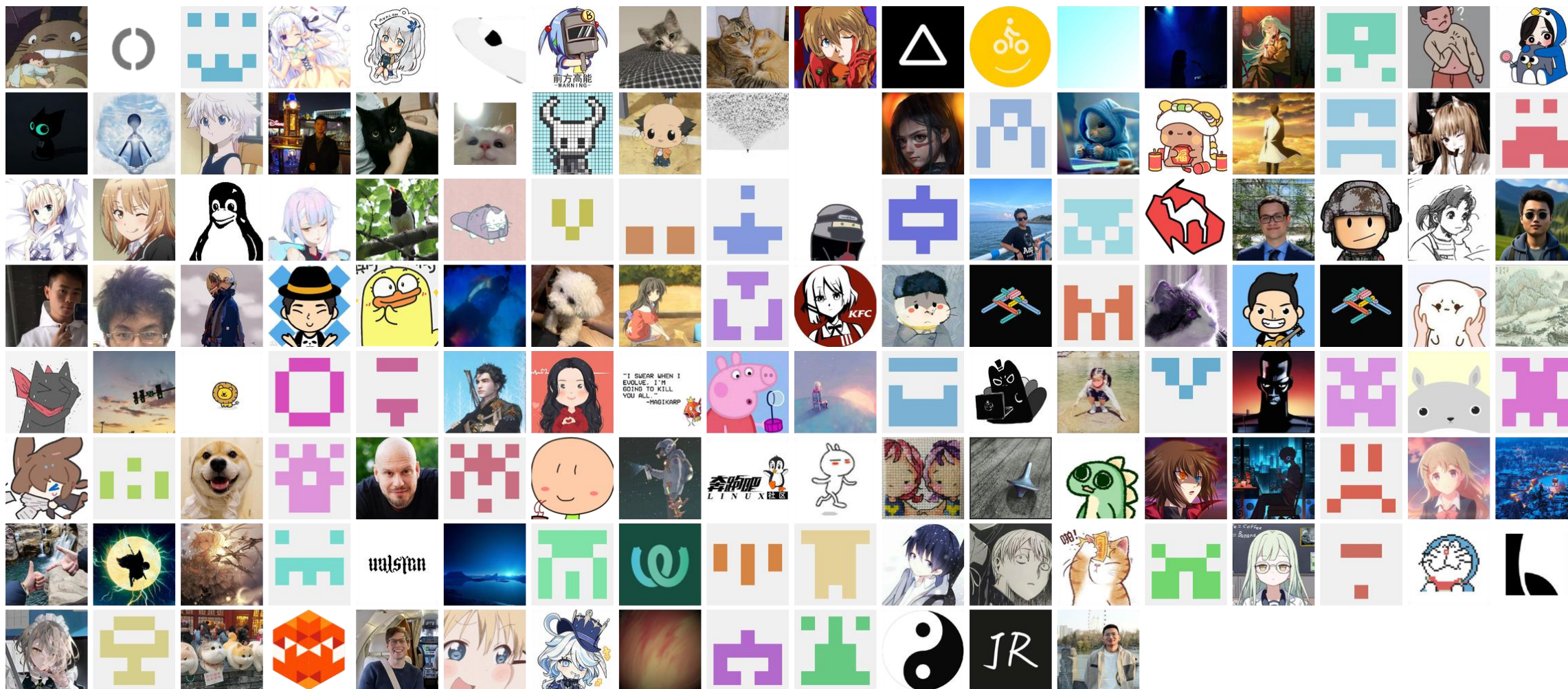
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# One of the Most Popular Open-Source Chip Projects

- **GitHub: >6700 Stars, > 800 Forks**

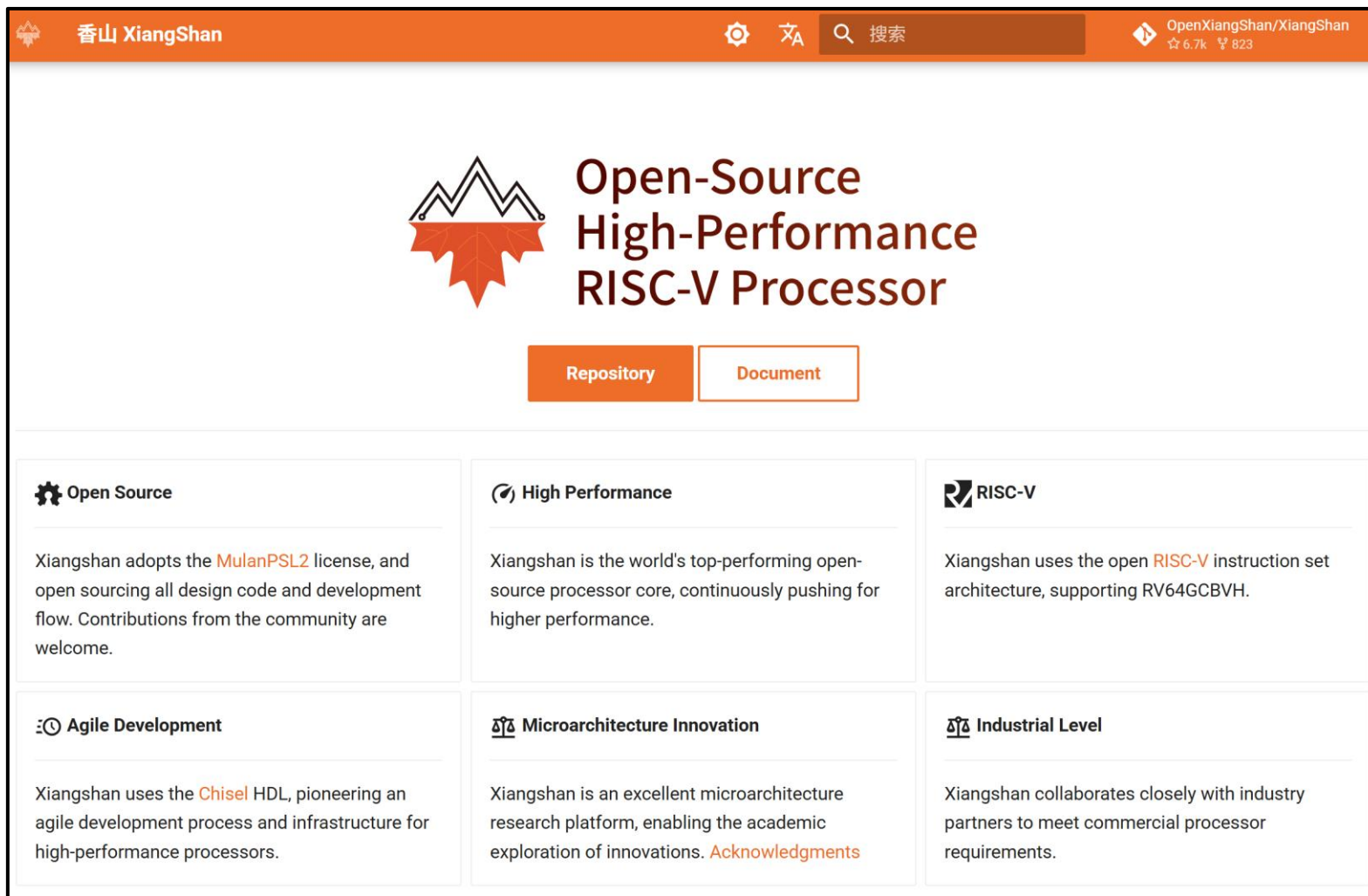




i50 i

# XiangShan Open-Source Community Website

- In 2024, XiangShan launched its own domain and website [XiangShan.cc](https://XiangShan.cc)



# XiangShan Open-Source Community Conference

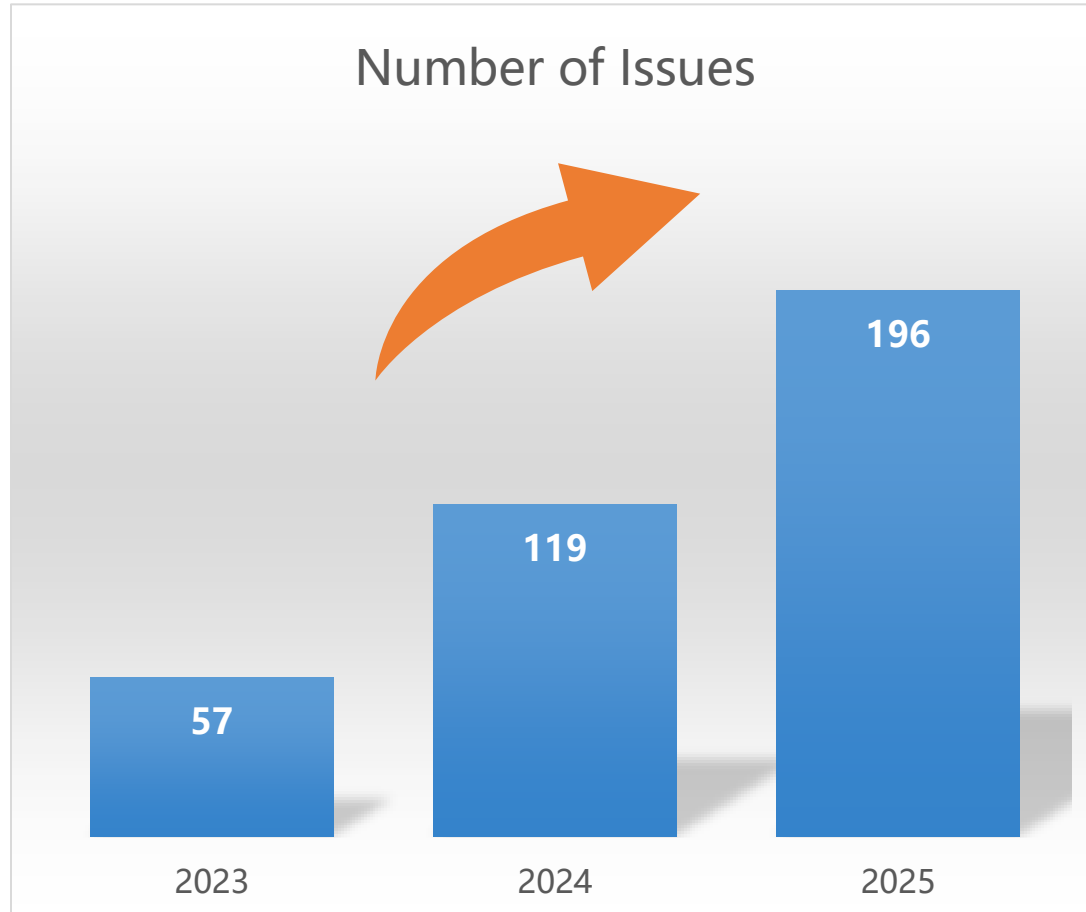


The 4<sup>th</sup> RISC-V Summit China  
2024 Hangzhou, China



The 5<sup>th</sup> RISC-V Summit China  
2025 Shanghai, China

# GitHub Issue & Discussion



- Time to First Response for Issues
  - Average: 25 hours
  - Median: 13 hours
- free to create a new issue on GitHub

# Acknowledgments in XiangShan

- A list outlines 32 techniques used in the XiangShan RTL codes
- <https://docs.xiangshan.cc/acknowledgments/>



## Multi Port Data Cache

[1] Gurindar S. Sohi, and Manoj Franklin. "High-bandwidth data memory systems for superscalar processors." 4th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS). 1991. [\[RTL Codes\]](#)

## Data Replacement

[1] Aamer Jaleel, Kevin B. Theobald, Simon C. Steely, and Joel Emer. "High performance cache replacement using reference interval prediction (RRIP)." 37th Annual International Symposium on Computer Architecture (ISCA). 2010. [\[RTL Codes\]](#)

## Data Prefetch

[1] Jean-Loup Baer, and Tien-Fu Chen. "An effective on-chip preloading scheme to reduce data access penalty." ACM/IEEE Conference on Supercomputing. 1991. [\[RTL Codes\]](#)

[2] Stephen Somogyi, Thomas F. Wenisch, Anastassia Ailamaki, Babak Falsafi and Andreas Moshovos. "Spatial memory streaming." 33rd International Symposium on Computer Architecture (ISCA). 2006. [\[RTL Codes\]](#)

[3] Santhosh Srinath, Onur Mutlu, Hyesoon Kim, and Yale N. Patt "Feedback directed prefetching: Improving the performance and bandwidth-efficiency of hardware prefetchers." IEEE 13th International Symposium on High Performance Computer Architecture (HPCA). 2007. [\[RTL Codes\]](#)

```
master XiangShan / src / main / scala / xiangshan / mem / prefetch / SMSPPrefetcher.scala

Siudya feat(dfx): integrate dfx components (#4312) ✓

Code Blame 1359 lines (1207 loc) · 58.6 KB · ⓘ

1 /*****
2 * Copyright (c) 2024 Beijing Institute of Open Source Chip (BOISC)
3 * Copyright (c) 2020-2024 Institute of Computing Technology, Chinese Academy of Sciences
4 * Copyright (c) 2020-2021 Peng Cheng Laboratory
5 *
6 * XiangShan is licensed under Mulan PSL v2.
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8 * You may obtain a copy of Mulan PSL v2 at:
9 * http://license.coscl.org.cn/MulanPSL2
10 *
11 * THIS SOFTWARE IS PROVIDED ON AN "AS IS" BASIS, WITHOUT WARRANTIES OF ANY KIND,
12 * EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO NON-INFRINGEMENT,
13 * MERCHANTABILITY OR FIT FOR A PARTICULAR PURPOSE.
14 *
15 * See the Mulan PSL v2 for more details.
16 *
17 *
18 * Acknowledgement
19 *
20 * This implementation is inspired by several key papers:
21 * [1] Stephen Somogyi, Thomas F. Wenisch, Anastassia Ailamaki, Babak Falsafi and Andreas Moshovos. "[Spatial memory
22 * streaming.](https://doi.org/10.1109/ISCA.2006.38)" 33rd International Symposium on Computer Architecture (ISCA).
23 * 2006.
24 *
25 *
26 package xiangshan.mem.prefetch
27
28 import org.chipsalliance.cde.config.Parameters
29 import chisel3._
30 import chisel3.util._
31 import utils._
32 import utility._
```

# Biweekly Report in English

- our recent progress and performance data
- <https://docs.xiangshan.cc/zh-cn/latest/blog/category/biweekly-en/>

## Biweekly-en

2025年9月29日 · 分类于 [Biweekly-en](#) · 需要 5 分钟阅读时间

[\[XiangShan Biweekly 86\] 20250929](#)

Welcome to XiangShan biweekly column! Through this column, we will regularly share the latest development progress of XiangShan. We look forward to your contribution.

This is the 86th issue of the biweekly report.

We are very pleased to share two pieces of news with you.

On September 20, the XiangShan team won the first Open Source Contribution Award from the CCF Architecture Committee. This collective award holds special significance for the XiangShan team—it represents recognition and support from our academic peers for the open-source processor and the team itself, laying the foundation for XiangShan to have a broad impact. The XiangShan team will continue to move forward, step by step, striving to keep XiangShan alive for 30 years!

On September 22, Innosilicon released the "Fenghua 3" full-featured GPU. The "Fenghua 3" GPU successfully integrated the XiangShan "Nanhu" processor IP core, which is performance-competitive with the ARM Cortex-A76, as its high-performance on-chip main control CPU. This integration marks a new phase in the industrial application of open-source high-performance CPU IPs and signifies that RISC-V can carve out a path different from the traditional ARM model.

We believe that open-source chips do not equate to low performance or low quality. Open source will profoundly change the cost structure of chip development, providing a new paradigm for chip design in the industry.





# User Guide

- For Software Developers and Hardware Integrators
- <https://docs.xiangshan.cc/projects/user-guide/en/latest/>



## XiangShan Open-Source Processor User Guide

Applicable to Kunminghu V2R2

e27508a  
2025 年 9 月 1 日



# Design Document

- Detailed Docs on Microarchitecture and Modules
- <https://docs.xiangshan.cc/projects/design/en/latest/>



## XiangShan Open-Source Processor Design Document

Applicable to Kunminghu V2R2

52fc49d  
2025 年 7 月 15 日

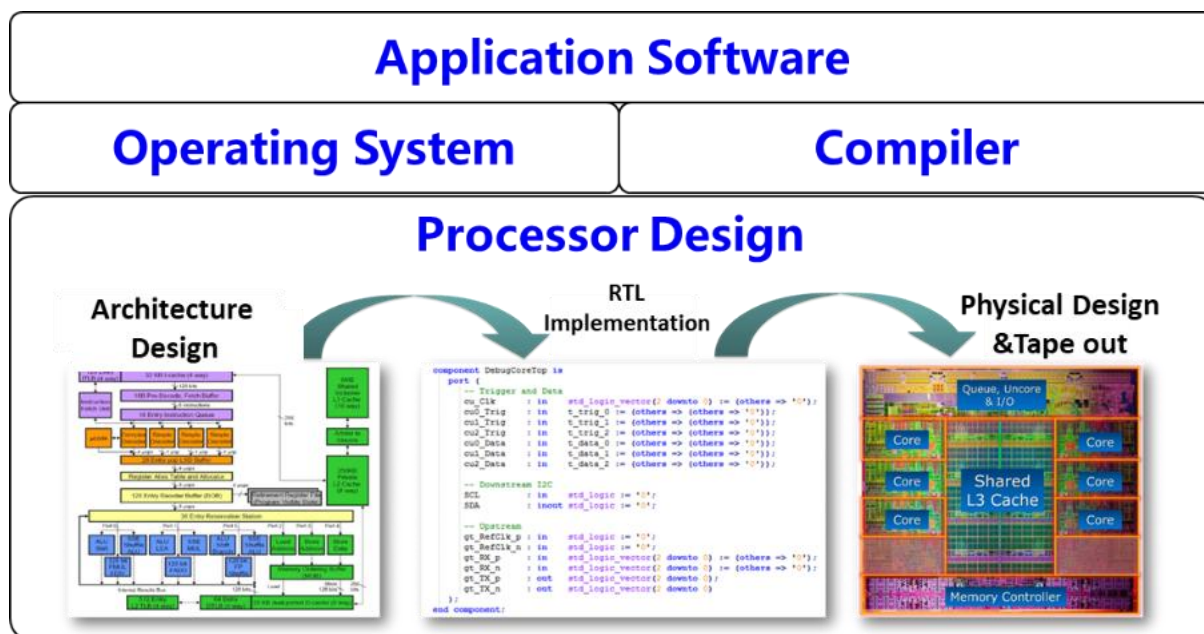


# The One Student One Chip (OSOC) Initiative



OSOC Docs

- **Learning-by-Doing:** Teach undergraduates to build real chips
- Build the full stack of the computer system, **from HW to SW**
- Involves thousands of students in China and extends its impact overseas



The first microprocessor designed in Kazakhstan



# UnityChip: A Croudsourcing Platform for Chip Verification

- **Features:** Involve software engineers

- Enable crowdsourcing
- Support multi-languages
- Be compatible with UVM

- **Effectiveness**

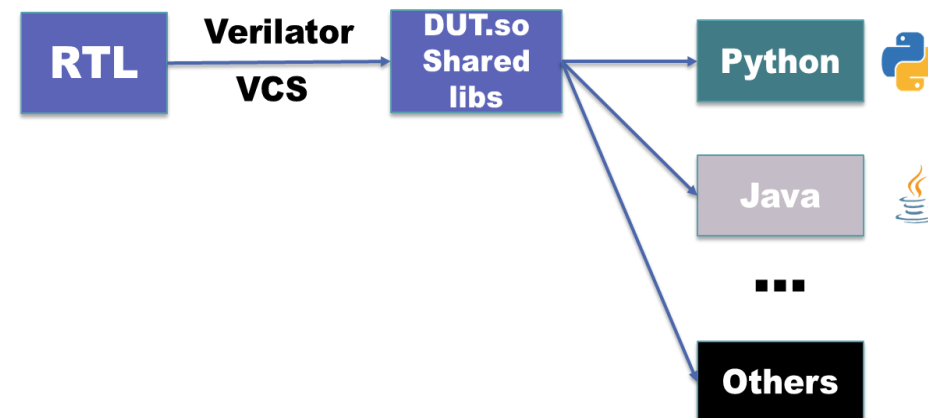
- 5 undergraduate students found **10 bugs** in 2 months

- **Easy-to-Use**

- Students not familiar with Linux and Python
- Learn to use tools in **5 days**
- Start adding test cases after another **10 days**

- **UnityChip Competition for XiangShan**

- Task: Unit Test for IFU, BPU, ITLB, etc.
- Scan the QR code for more information



GitHub Link

## Together for a Shared Future

- Feel free to contact us through email or file issues on GitHub!
  - [all@xiangshan.cc](mailto:all@xiangshan.cc)
  - <https://github.com/OpenXiangShan/XiangShan>



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